

Exploring the Differential Impact of Repealing the ACA Individual Mandate by Health Status

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Abstract

Health insurance mandates are designed to mitigate adverse selection by encouraging enrollment among relatively healthy individuals. Yet, little is known about how removing these incentives affects relatively healthy and sick consumers. This paper examines the differential impact of repealing the Affordable Care Act (ACA) individual mandate on insurance participation, Marketplace premiums, out-of-pocket premium spending, and consumer surplus. I exploit the 2019 repeal of the federal mandate penalty using a two-way fixed effects (TWFE) design that compares states without their own mandates to states that retained mandate policies. I find that the repeal increased the probability of becoming newly uninsured by 0.56 percentage points (approximately 27 percent relative to the pre-reform mean), with nearly all of the increase driven by relatively healthy consumers, while insurance participation among relatively sick consumers remained largely unchanged. Benchmark Marketplace premiums for a 27-year-old enrollee increased by approximately \$739, while average out-of-pocket premium spending decreased by \$284.60 because the ACA premium subsidies offset much of the increase in benchmark premiums. These findings demonstrate that ACA premium subsidies substantially alter the distributional consequences of adverse selection by limiting the pass-through of higher Marketplace premiums to consumers who remain insured.

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1 Introduction

A major rationale for insurance mandates is its potential, in conjunction with other policies, to help mitigate adverse selection in insurance markets ([Akerlof, 1970](#); [Rothschild and Stiglitz, 1976](#); [Wilson, 1977](#)). However, critics argue that mandating coverage constrains individual choice and imposes financial burdens, particularly on low-income individuals ([Young, 2009](#); [Lurie et al., 2021](#); [Soni, 2022](#)). The absence of a mandate likewise has consequences. Proponents of repealing the mandate contend that it relieves low-risk individuals of the premium costs associated with being insured or of tax penalties for not having coverage. In contrast, opponents argue that removing the mandate can lead to an increasingly high-risk risk insurance pool, placing upward pressure on premiums and healthcare expenditures ([Saltzman, 2019](#)). In addition, those who drop coverage are now exposed to more financial risk due to an unexpected health care expense.

Despite these debates, relatively little is known about how repealing insurance mandates affects consumer coverage decisions and healthcare spending by health status. In this paper, I estimate the impact of the 2019 repeal of the Affordable Care Act (ACA) individual mandate on insurance coverage and expenditure. If the insurance mandate imposes financial costs on individuals with relatively low demand for coverage, then its repeal may yield short-term financial gains for them as they drop coverage. On the other hand, increased exposure to health shocks and out-of-pocket financial risk may offset those gains to some extent. In addition, those with relatively high demand for coverage (i.e. the sick) now face higher premiums due to adverse selection. Understanding how the removal of insurance mandates affects consumers with different levels of health status is therefore important for both economists and policymakers interested in designing welfare-improving insurance policies.

Before the repeal, Americans with household incomes above 138% of the federal poverty line were required to obtain coverage or pay a tax penalty. Previous work attempting to isolate the effect of the mandate apart from the other components of the ACA found that the mandate increased insurance coverage by roughly 2.5 percentage points in 2016 ([Lurie et al., 2021](#)). However, the ACA individual mandate was repealed by the "Tax Cuts and Jobs Act" of 2017, which set the

penalty for not having health insurance to zero dollars, effective from January 1, 2019. Following the federal repeal, three states (Massachusetts, New Jersey, and Washington, D.C.) implemented state-level mandates on January 1, 2019, leaving the other 48 states without an insurance mandate (Soni, 2022).¹ This variation across states provides a natural experiment to study how the repeal of the ACA individual mandate shapes insurance coverage and healthcare spending by health status using a two-way fixed effects model.

To guide my empirical analysis, I build on the conceptual framework developed by Geruso et al. (2023a), which distinguishes between insurance demand at the extensive margin (whether to purchase insurance) and the intensive margin (the generosity of insurance coverage). Their framework predicts that changes in mandate policy can affect both the decision to purchase insurance and the choice of insurance generosity, with implications for equilibrium premiums and welfare. My analysis applies this framework to the repeal of the ACA individual mandate in order to generate testable predictions about changes in insurance coverage and out of pocket premium payments driven by the repeal.

I test these predictions using a sample of non-elderly adults from the 2015–2019 Current Population Survey Annual Social and Economic Supplement (CPS ASEC). The CPS ASEC reports both current and past-year insurance coverage, allowing me to examine how consumers adjusted their coverage decisions following the repeal. It also contains information on out-of-pocket premium spending and self-reported health status, enabling me to estimate how the repeal affected insurance coverage and expenditure by health status.

My analysis addresses three related questions. First, I examine whether the repeal of the ACA individual mandate increased adverse selection. I find that repeal increased the probability of becoming newly uninsured by 0.56 percentage points which is approximately 27% relative to pre-reform mean among the full sample. The increase was driven entirely by relatively healthy individuals, while coverage among sick enrollees remained unchanged.

Second, I examine the effect of the repeal on consumers' out-of-pocket premium expenditure.

¹Figure 1 presents the variation in insurance mandate penalty after the repeal in 2019

I find that out-of-pocket premium spending declined by \$284.60 (approximately 18%) relative to the pre-reform mean for the full sample. Again, I find that the decline is concentrated among relatively healthy consumers. Although out-of-pocket premium spending also decreases by a similar magnitude among higher-risk consumers, the estimate is not statistically distinguishable from zero.

Third, I estimate the impact of the repeal on benchmark Marketplace premium. I find that the repeal increased the annual benchmark Marketplace premium by \$738.93 for a 27-year-old individual in 2019 equivalent to approximately 21% of the pre-repeal mean. The corresponding increase was \$1,361.46 for a 50-year-old individual and \$3,341.60 for a 30-year-old couple. These estimates indicate that the repeal substantially increased the unsubsidized (i.e. gross) cost of insurance in the individual market. The larger dollar increases for older adults and couples are consistent with the ACA's age-rating structure and their higher expected medical costs.

This paper makes two contributions to the literature. First, it contributes to the growing literature on the effects of the ACA individual mandate repeal on insurance coverage ([Zhang, 2021](#); [Soni, 2022](#)). While existing studies document that repealing the mandate increases the uninsured rate, they provide limited evidence on which consumers leave the insurance market. I fill this gap by showing that the increase in the newly uninsured is concentrated among relatively healthy consumers, whereas coverage among relatively sick consumers remains largely unchanged.

Second, this paper contributes to the literature on adverse selection in health insurance markets ([Einav et al., 2010](#); [Einav and Finkelstein, 2011](#); [Castaneda and Marton, 2013](#)).² While this literature has shown that adverse selection raises equilibrium premiums and reduces welfare, there is comparatively little causal evidence on how these market-level effects translate into consumers' out-of-pocket premium spending. Distinguishing between equilibrium premiums and the premiums consumers actually pay is particularly important in the presence of premium subsidies, which can insulate insured individuals from increases in market premiums. I provide causal

²For a more recent literature, see [Handel et al. \(2015\)](#), [Hackmann et al. \(2015\)](#), [Geruso and Layton \(2017\)](#), [Marton et al. \(2017\)](#), [Saltzman \(2018\)](#), [Saltzman \(2019\)](#), [Lurie et al. \(2021\)](#), [Shepard \(2022\)](#), [Geruso et al. \(2023a\)](#), [Geruso et al. \(2023b\)](#), [Kreider et al. \(2024\)](#), and [Peterson \(2025\)](#).

evidence that, although the repeal of the ACA individual mandate induced the exit of relatively healthy consumers and intensified adverse selection, out-of-pocket premium payments declined rather than increased. I argue that this pattern reflects the ACA’s premium subsidy structure, which limits premium pass-through for eligible households with incomes between 138% and 400% of the federal poverty line. More broadly, my findings show that premium subsidies can substantially attenuate the extent to which increases in equilibrium premiums are passed through to consumers following mandate repeal.

The rest of the paper is structured as follows. Section 2 focuses on the conceptual framework describing the interaction between adverse selection, individual mandate penalty and its repeal. In Section 3, I provide the background on my empirical setting from the ACA individual mandate. The Section 4 describes the data, and Section 5 discusses the empirical strategy. Section 6 presents the results, and Section 7 concludes the paper.

2 Conceptual Framework

I begin by adapting the conceptual framework of [Geruso et al. \(2023a\)](#) to illustrate how an individual mandate affects insurance demand in a market with adverse selection. Their model allows consumers to choose among a more generous plan, Q_H , a less generous plan, Q_L , and an outside option of remaining uninsured, Q_U .

Imposing a mandate penalty raises the cost of Q_U and induces relatively healthy, low-cost consumers at the $Q_L - Q_U$ margin to enroll in Q_L . Their entry lowers the average cost, and hence the equilibrium price of Q_L . The resulting reduction in the price of Q_L then induces some relatively higher-risk consumers at the $Q_H - Q_L$ margin to switch from Q_H to Q_L , shifting demand away from the more generous plan, as illustrated in Figure 2. Thus, the mandate generates responses along both the extensive and intensive margins: it increases insurance coverage among relatively healthy consumers while inducing some higher-risk consumers to substitute toward less generous coverage. These cross-margin responses imply that the effects of a mandate cannot

be inferred from its direct effect on insurance participation alone.

These effects operate in reverse when the mandate penalty is removed. Relatively healthy, low-cost consumers at the $Q_L - Q_U$ margin move from plan Q_L into uninsurance, worsening the risk composition of plan Q_L and raising its equilibrium price. The resulting increase in the price of Q_L induces some relatively higher-risk consumers at the $Q_H - Q_L$ margin to switch toward the more generous plan Q_H . This intensive-margin response changes the allocation of higher-risk consumers across plans and thus the equilibrium price of each plan.

This framework motivates my empirical analysis as it yields several testable predictions. Following the removal of the mandate, relatively healthy consumers on the margin of having coverage or not should be more likely to return to non-coverage. Higher-risk consumers are predicted to remain insured in the more generous plan.³ The effect on out-of-pocket premium spending among relatively healthy consumers is ambiguous. Lower-risk consumers who remain insured may face higher equilibrium premiums in the less generous plan if some healthy consumer drop coverage, however those premiums could also fall as consumers on the margin between the more and less generous plans move back into the more generous plan. In addition, those who become uninsured no longer make premium payments. Higher-risk consumers, who are more likely to remain insured in the more generous plan, may face slightly lower premiums due to consumers on the margin moving from the less generous to the more generous plan.

3 Background on the ACA Individual Mandate

The Affordable Care Act (ACA), signed into law in March 2010, represents the largest overhaul of the U.S. healthcare system since the launch of Medicare and Medicaid in the mid-1960s (Frean et al., 2017). It was primarily designed to expand health insurance coverage in the United States by implementing market reforms, individual mandates, premium subsidies, health insurance exchanges, and Medicaid expansions, while also aiming to address the escalating cost of healthcare

³Appendix A.1 provides a more detailed illustration of these predictions using the institutional settings of the ACA individual mandate.

(Gruber, 2011; Courtemanche et al., 2017).

The major components of the ACA, including the individual mandate, were introduced in January 2014. The mandate requires Americans with household incomes above 138% of the FPL to buy health coverage or face a tax penalty (Gruber, 2011; Courtemanche et al., 2017).⁴ The goal of the ACA individual mandate was to mitigate adverse selection by encouraging healthier individuals to buy coverage (Fiedler, 2020; Soni, 2022). However, exemptions were granted to specific groups, including individuals with an income below the tax filing threshold and those who did not have access to a health insurance plan costing less than 8% of their annual income in 2014 and 8.05% in 2015 (Saltzman, 2019). In addition, exemptions were granted to those who would have qualified for Medicaid but lived in a non-expansion state (Saltzman, 2019).

Critics of the mandate provision consider it an undue infringement on their freedom and a financial burden on low-income Americans in spite of the subsidies built into the ACA (Young, 2009; Lurie et al., 2021; Soni, 2022). In 2017, the United States Congress passed the "Tax Cuts and Jobs Act", which effectively repealed the federal individual mandate by reducing the penalty for not having health insurance to zero dollars, starting from January 1, 2019. Following the federal repeal, three states (Massachusetts, New Jersey, and the District of Columbia) implemented state-level mandates in January 1, 2019.⁵ Thus, uninsured individuals living in states with an individual mandate are still penalized, while uninsured residents in states without such a mandate are not penalized.⁶ This variation across states provides a natural experiment to study the effects of the repeal on insurance coverage and out-of-pocket premium spending by health status.

⁴In 2014, the penalty was the greater of 1% of household income, capped at the national average annual premium for a Bronze plan, or a flat fee of \$95 per adult and \$47.5 per child, with a maximum of \$285 per family. By 2015, the penalty was the greater of either \$325 per adult or 2% of household income capped at \$975 per family; and in 2016, the penalty was the greater of either \$695 per adult or 2.5% of household income capped at \$2085 (Courtemanche et al., 2017; Saltzman, 2018).

⁵The following year, three additional states (California, Rhode Island, and Vermont) joined the list of states with health insurance mandates (Soni, 2022).

⁶Table B.1 in the appendix summarizes states with and without mandates in 2019.

4 Data Description

I combine several data sources to estimate the effects of the ACA individual mandate repeal. I use the Current Population Survey Annual Social and Economic Supplement (CPS ASEC) to measure insurance coverage and consumers' out-of-pocket premium payments and the Robert Wood Johnson Foundation (RWJF) Health Insurance Exchange data to measure market premiums. In supplementary analysis, I use the Behavioral Risk Factor Surveillance System (BRFSS) to examine changes in healthcare utilization. Together, these data sources allow me to study the effects of mandate repeal on insurance participation, market premiums, consumer premium payments, and healthcare utilization by health status.

For my main analysis, I use data from the 2015–2019 CPS ASEC (Flood et al., 2023). Conducted annually between February and April, the CPS ASEC provides detailed information on respondents' health insurance coverage, state of residence, and socioeconomic characteristics. Importantly, the survey reports both current and prior-year insurance coverage, allowing me to identify individuals who were previously insured but are uninsured at the time of the interview. This feature enables me to examine changes in insurance coverage following the repeal of the ACA individual mandate.

I construct "Newly Uninsured", an indicator equal to one if a respondent was insured in the previous year but uninsured at the time of the interview, and zero otherwise. Similarly, I define "Newly Insured" as an indicator equal to one if a respondent was uninsured in the previous year but insured at the time of the interview, and zero otherwise.⁷ I also construct an indicator for "good health" using respondents' self-reported health. The indicator equals one for respondents reporting excellent, very good, or good health and zero for those reporting fair or poor health. I refer to these groups as relatively healthy and sick consumers, respectively. The 2015–2019 CPS ASEC also reports respondents' out-of-pocket premium payments, which I use to estimate the effect of mandate repeal on consumer premium spending by health status. Finally, the survey provides detailed demographic and socioeconomic characteristics, including state of residence,

⁷I use these measures to validate the findings of Soni (2022) (see Appendix Table ??).

age, sex, marital status, educational attainment, employment status, and race and ethnicity.

There are many advantages associated with using the CPS ASEC data. First, it is less prone to insurance misreporting. Instead of asking respondents a series of yes/no questions, the survey first asks whether they have coverage, then asks about their general sources of insurance, and finally asks follow-up questions about their specific coverage (Pascale et al., 2019). Second, the CPS-ASEC survey on insurance measures has been independently validated as more accurate compared to other notable health insurance surveys, which tend to overestimate the uninsured population (Pascale et al., 2019; Soni, 2022). These features give confidence in the reliability of the insurance measures used in my analysis.

In addition, I use 2015–2019 Health Insurance Exchange data from the RWJF to examine how mandate repeal affected market premiums. This dataset includes detailed information on premiums offered by insurers across the 50 states and the District of Columbia, including state-year benchmark premiums before the application of ACA premium subsidies. These data allow me to distinguish between changes in market premiums from changes in the out-of-pocket premium payments observed in the CPS ASEC. I exploit variation in premiums over time and across states to compare premium trends between states that retained an individual mandate and those that did not following the federal repeal. Combining these data with the CPS ASEC therefore allows me to examine whether mandate repeal increased market premiums even when consumers’ out-of-pocket premium payments did not increase.

5 Empirical Strategy

This section describes the empirical methodology used throughout the paper. I begin by outlining the identification strategy used to estimate the causal effects of the repeal of the ACA individual mandate. I then present the two-way fixed effects model used to estimate the effects on the outcomes of interest, followed by a two-part model for outcomes with a large share of zero observations. Finally, I discuss the robustness checks and identification assumptions used to

assess the validity of the empirical estimates.

5.1 Identification Strategy

To identify the causal effects of mandate repeal on insurance coverage and out-of-pocket premium spending by health status, I restrict the sample to non-elderly adults with household incomes between 138% and 400% of the FPL. I exclude adults aged 65 and older because Medicare eligibility largely insulates them from the effects of the individual mandate. Individuals younger than 19 are excluded because health insurance decisions for this group are typically made by parents or guardians rather than by the individuals themselves. I also exclude individuals with household incomes below 138 percent of the FPL because they are generally eligible for Medicaid and therefore face different insurance incentives than those purchasing private coverage.

Finally, I exclude individuals with household incomes above 400% of the FPL because they are generally ineligible for ACA Marketplace premium subsidies during the study period. Moreover, higher-income households are more likely to obtain coverage through employer-sponsored insurance rather than the individual Marketplace. Restricting the sample to subsidy-eligible households therefore yields a more homogeneous population whose insurance decisions are directly shaped by the ACA Marketplace and the individual mandate.

5.2 Two-Way Fixed Effects

I examine the variation in mandates across states to estimate the causal effects of repealing the ACA individual mandate. I estimate the following baseline model using the two-way fixed effects (TWFE) framework:

$$y_{ist}^k = \alpha + \beta 1[\text{Treatment}_s \times \text{post2019}_t] + \gamma X_{ist} + \lambda_s + \tau_t + \epsilon_{ist} \quad (1)$$

where y_{ist}^k denotes the outcome for individual i living in state s in year t . The outcomes include out-of-pocket premium spending, the probability of becoming newly uninsured, and the

probability of becoming newly insured. $Treatment_s$ is an indicator equal to one for individuals residing in states that did not implement a state-level individual mandate and zero otherwise. $post2019_t$ is an indicator equal to one for observations in 2019 and later, and zero otherwise. X_{ist} is a vector of individual demographic and socioeconomic characteristics, including age, sex, race and ethnicity, marital status, educational attainment, household size, and employment status. The model also includes state fixed effects, λ_s , and year fixed effects, τ_t . Finally, ϵ_{ist} denotes the idiosyncratic error term.

Using the TWFE framework, I control for individuals' state of residence to account for unobserved differences across states. I also control for the year of the interview to account for variations in nationwide trends in insurance coverage. The parameter β measures the effect of eliminating the federal insurance mandate on the outcomes of interest. I cluster the standard errors at the state level. The estimates include CPS ASEC survey weights.

5.3 Two-part Model

Similar to other healthcare spending data, out-of-pocket premium spending for health insurance coverage is skewed because (i) out-of-pocket premium spending cannot be negative, (ii) there is a concentration of values at zero for both insured and uninsured individuals, and (iii) older adults are likely to pay higher premiums than younger adults. Given the presence of mass zeroes in the premium expenditure distribution, I employ a two-part model approach developed by [Belotti et al. \(2015\)](#) to estimate the effects of repealing the ACA individual mandate on out-of-pocket premium spending. This model effectively addresses the skewness associated with discrete-continuous outcomes, which is a common feature of medical expenditure data. The approach splits out-of-pocket premium spending into two components (extensive and intensive margin).⁸ It applies the basic principles of probability to estimate the parameters in the conditional mean function $E(y|x) = (y > 0|x) \times E(y|y > 0, x)$.

The first part of the model estimates the extensive margin of out-of-pocket premium-spending.

⁸The extensive margin estimates the effects of the repeal on the premium spending of insured and uninsured consumers. On the other hand, the intensive margin only focuses on insured individuals.

In other words, the outcome represents having any spending vs none. I use the logit model to estimate the probability of incurring out-of-pocket premium expenditure $pr(Y_{ist} > 0|x) = \frac{1}{1+e^{-\beta X}}$. Where Y_{ist} is a binary outcome equal to 1 if out-of-pocket premium expenditure for an individual i residing in state s in year t is positive and zero otherwise. Here, X is a vector of covariates that includes the interaction term.

In the second part, I use the Generalized Linear Model (GLM) developed by [McCullagh \(2019\)](#), which provides more consistent estimates than the OLS method commonly used in previous studies. Prior works that used OLS, applied log transformations to the data, a practice now generally discouraged due to heteroscedasticity issues according to [Manning and Mullahy \(2001\)](#). The GLM is considered robust because its estimates consistently outperform those obtained from OLS regardless of whether transformations are applied. Thus, I use the GLM models with a log-link function and a gamma distribution to estimate the effect of the repeal on out-of-pocket premium spending, conditional on incurring positive costs, as shown in the following equation: $E(Y_{ist}|Y_{ist} > 0, X) = e^{X\theta}$. Where Y_{ist} represents the out-of-pocket premium spending incurred by individual i residing in state s in year t , and X is a vector of covariates as specified in the first part of the analysis.

5.4 Validation of Identifying Assumptions

The idea behind the TWFE model, as expressed in Equation 1, is based on two key assumptions. First, my outcome variables of interest would have trended similarly in the treatment and control states in the absence of the policy change. Although I cannot test the counterfactual directly, I can compare trends in the pre-period by estimating Equation (2). The absence of differential trends in the pre-period suggests evidence that my identifying assumptions hold.

$$y_{ist}^k = \alpha + \sum_{\substack{j=2015 \\ j \neq \{2018\}}}^{2019} \beta_j 1[\text{Treatment}_{sj}] + \gamma X_{ist} + \lambda_s + \tau_t + \epsilon_{ist} \quad (2)$$

Where $Treatment_{sj}$ is a set of event-time indicators identifying the years before and after

the repeal of the federal individual mandate. The event-time variable, j , indexes years relative to the repeal, with 2018 serving as the omitted reference year. Values of j after 2018 represent post-repeal periods (lags), whereas values before 2018 represent pre-repeal periods (leads). The coefficient β_j measures the effect of mandate repeal on the outcome of interest relative to the reference year. All other variables are defined as in Equation (1).

Second, I assume that the outcomes in both the treatment and control states should not undergo any further differential changes during the post-treatment period. The validity of this assumption was tested by conducting two falsification tests using Equation (1) across two groups of adults with no incentive to drop insurance coverage following the mandate repeal. I regress Equation (1) on the elderly adults in my sample (who were eligible for Medicare before and after the repeal and are unaffected by the mandate repeal). I also show the estimates for low-income individuals with household incomes below 138% of the federal poverty line (who are eligible for Medicaid and were exempted from the ACA mandate penalty before and after the policy change). These groups are less likely to be affected by the mandate repeal. Therefore, a significant causal effect of the repeal on my outcome variables of interest for either of these two groups would raise serious concerns about the interpretation of my primary results if there are any simultaneous changes in other policies.

I conduct several robustness checks to assess the validity of the main results. First, I re-estimate equation (1) using a broader sample that includes individuals with household incomes below 138% of the FPL. Second, because the treatment group is substantially larger and more geographically diverse than the control group (Massachusetts, New Jersey, and the District of Columbia), I restrict the sample to Northeastern states to improve the comparability of treated and control states. Finally, I re-estimate equation (1) excluding observations from 2018. This robustness check accounts for the possibility that the announcement of the Tax Cuts and Jobs Act in late 2017 may have affected insurance decisions before the federal mandate penalty was reduced to zero in 2019 (Fung et al., 2019; Soni, 2022).

Finally, I examine whether the effects of mandate repeal vary across population subgroups

by estimating the model separately for men and women, parents and childless adults, married and unmarried individuals, age groups, employment status, and educational attainment. I also conduct a series of placebo tests using outcomes that should not be directly affected by the repeal, including labor force participation, and self-employment. Finding no significant effects on these placebo outcomes would strengthen the interpretation that the estimated effects reflect changes induced by mandate repeal rather than unobserved confounding factors.

6 Results and Discussion

In this section, I present my empirical findings. Using both graphs and regression analysis, I show the impact of repealing the ACA individual mandate on insurance coverage, and out-of-pocket spending, and Marketplace premiums.

6.1 Baseline Estimates of the Effects of Repealing the ACA Mandate

I begin by presenting descriptive statistics for the 2015–2019 CPS ASEC sample in Table 1. The table reports the means of the outcome variables and respondents’ demographic characteristics before and after the repeal of the ACA individual mandate for both the treatment and control states. The demographic characteristics are broadly similar across the two groups prior to the repeal, suggesting that the treatment and control samples are comparable along observable characteristics. Column 5 illustrates that among the full sample there are statistically significant reductions in the likelihood of being newly insured and in out-of-pocket premium spending associated with the repeal. There is also a statistically significant increase in the likelihood of becoming newly uninsured and in Marketplace premiums.

Figure 3 plots trends in individual Marketplace enrollment using open enrollment data from the Centers for Medicare and Medicaid Services (CMS).⁹ The solid blue line shows total Marketplace enrollment, and the vertical line marks the repeal of the ACA individual mandate. Enroll-

⁹CMS reports annual open enrollment statistics for the federal Marketplace and state-based health insurance exchanges.

ment declines very modestly between 2018 and 2020, a pattern that is consistent with the repeal of the individual mandate following the Tax Cuts and Jobs Act of 2017. Enrollment begins to recover after 2020, likely reflecting expanded enrollment initiatives and temporary policy changes introduced during the COVID-19 pandemic. For completeness, the figure presents enrollment trends from 2015 through 2026. However, the empirical analysis is restricted to 2015 through 2019 to avoid the change in the way the CPS measured insurance between 2014 and 2015 and the potential confounding effects associated with the COVID-19 pandemic starting in 2020.

Table 2 reports the estimated effects of repealing the ACA individual mandate on health insurance coverage using equation (1) for the full sample and separately for the healthy and sick sub-samples. Columns 1–3 present the estimated effects on the probability of becoming newly uninsured. All specifications include individual demographic and socioeconomic controls. The repeal increased the probability of becoming newly uninsured by 0.56 percentage points ($p < 0.01$) in the full sample, equivalent to approximately 27% of the pre-reform mean. This estimate is close to the 0.5 percentage points increase reported by [Soni \(2022\)](#). Stratifying the sample by health status shows that the increase in uninsurance is driven entirely by relatively healthy consumers, whose probability of becoming newly uninsured increased by 0.57 percentage points following the repeal. By contrast, I find no statistically significant change among sick consumers, consistent with their greater willingness to pay for health insurance. These findings are consistent with the conceptual framework developed in Section 2, which predicts that relatively healthy consumers at the margin between purchasing a less generous plan and remaining uninsured are the most responsive to the removal of the mandate penalty.

Columns 4–6 report the estimated effects of mandate repeal on the probability of becoming newly insured. The repeal reduced the probability of becoming newly insured by 0.79 percentage points ($p < 0.01$), equivalent to approximately 34% of the pre-reform mean. This estimate is consistent with previous evidence that repealing the ACA individual mandate reduced insurance participation among individuals who would otherwise have enrolled under the mandate ([Soni, 2022](#)). Stratifying the sample by health status shows that the decline in new insurance enroll-

ment is concentrated among relatively healthy consumers, who became less likely to transition into coverage following the repeal. By contrast, I find no statistically significant effect among sick consumers, consistent with their greater willingness to pay for health insurance. Together with the results for newly uninsured individuals, these findings provide consistent evidence that repealing the individual mandate disproportionately affected insurance transitions among relatively healthy consumers, thereby intensifying adverse selection in the individual insurance market.

Table 3 reports the effects of mandate repeal on consumers' out-of-pocket premium spending using equation (1). Columns 1–3 present the estimates for annual out-of-pocket premium spending. For the full sample, the repeal reduced annual out-of-pocket premium spending by \$284.60, equivalent to approximately 18% of the pre-reform mean. Stratifying the sample by health status shows that this decline was driven primarily by relatively healthy consumers, whose annual premium spending fell by \$282.59. Although out-of-pocket premium spending also declined among sick consumers, the estimate in Column 3 is not statistically distinguishable from zero. These findings suggest that the reduction in out-of-pocket premium spending following mandate repeal was concentrated among relatively healthy consumers, while premium spending among sick consumers remained largely unchanged.

The conceptual framework developed in Section 2 yields different predictions for out-of-pocket premium spending across health-risk groups. For relatively healthy consumers, the overall effect is theoretically ambiguous. Following mandate repeal, the healthiest enrollees exit the insurance market and no longer pay insurance premiums. At the same time, the deterioration of the insured risk pool raises equilibrium premiums for those who remain enrolled, while changes in plan choice partially offset these increases. The estimates in Table 4 indicate that the net effect is negative, with relatively healthy consumers experiencing a significant decline in out-of-pocket premium spending.

For relatively sick consumers, the conceptual framework predicts a modest decline in out-of-pocket premium spending as some higher-risk enrollees shift toward more generous coverage,

improving the risk composition of the less generous plan. Consistent with this prediction, I estimate a reduction in out-of-pocket premium spending among sick consumers. However, the estimate is not statistically distinguishable from zero, suggesting that mandate repeal had little measurable effect on the premiums paid by consumers with greater healthcare needs.

Next, Columns 1–3 of Table 4 report the estimated effects of repealing the ACA individual mandate on Marketplace premiums using Equation (1). The repeal increased the annual benchmark premium by \$738.93 for a 27-year-old individual in 2019 equivalent to approximately 21% of the pre-repeal mean. The corresponding increase was \$1,361.46 for a 50-year-old individual and \$3,341.60 for a 30-year-old couple. These estimates indicate that the repeal substantially increased the unsubsidized (i.e. gross) cost of insurance in the individual market. The larger dollar increases for older adults and couples are consistent with the ACA’s age-rating structure and their higher expected medical costs. Overall, the results suggest that the exit of relatively healthy consumers following mandate repeal raised equilibrium premiums, consistent with the adverse-selection mechanism developed in the conceptual framework.

6.2 Event Study

I estimate event-study specifications for all outcomes using equation (2). Figure 4 reports the event-study estimates for the probabilities of becoming newly insured (Panel A) and newly uninsured (Panel B) using the 2015–2019 CPS ASEC. The estimates are presented separately for the full sample and for the healthy and sick sub-samples. All coefficients are normalized to zero in 2018, the year immediately preceding the repeal, so that each estimate is interpreted relative to the pre-repeal period. Across all samples, the pre-repeal coefficients are economically small and statistically indistinguishable from zero, indicating no evidence of differential pre-trends between the treatment and control states. These findings support the parallel-trends assumption and reinforce the validity of the empirical strategy.

Figure 5 reports the event-study estimates for out-of-pocket premium spending before and after the repeal of the ACA individual mandate. The estimates are reported separately for the

full sample and for the healthy and sick sub-samples. As in the insurance coverage analysis, all coefficients are normalized to zero in 2018, the year immediately preceding the repeal, and are therefore interpreted relative to the pre-repeal period. The pre-repeal coefficients are economically small and statistically indistinguishable from zero across all samples, indicating no evidence of differential pre-trends between the treatment and control states. Consistent with the event-study evidence for insurance coverage, these results support the parallel-trends assumption and reinforce the validity of the empirical strategy.

6.3 Heterogeneity in the Repeal of Individual Mandate Reform

Table 5 reports heterogeneity analyses for relatively healthy consumers using the CPS ASEC. Columns 1–2 present estimates for insurance transitions. I examine whether the effects of mandate repeal vary by sex, race and ethnicity, age, educational attainment, employment status, marital status, and parental status. I find that the effects of mandate repeal vary substantially across demographic groups. The largest responses are observed among unmarried individuals, childless adults, the self-employed, younger adults ages 19–26, and adults ages 60–64. These patterns are consistent with differences in insurance demand and reliance on the individual insurance market. By contrast, I find little evidence of differential responses among adults aged 27–59, Black and Hispanic consumers. Overall, the heterogeneity results reinforce the central finding of the paper that mandate repeal disproportionately reduced insurance participation among consumers with relatively weaker incentives to maintain insurance coverage.

Column 3 reports heterogeneity in the effects of mandate repeal on out-of-pocket premium spending. The decline in premium spending is more pronounced among women, married individuals, adults with children, individuals with a high school education or less, Hispanics and individuals in the “Other” race category, and employed workers who are not self-employed. The reduction is also largest among adults ages 27–49. These patterns suggest that the financial consequences of mandate repeal differed across demographic groups. One possible explanation is that some consumers responded to the repeal by selecting lower-premium coverage, while ACA

premium subsidies continued to shield eligible enrollees from increases in Marketplace premiums. By contrast, I find no statistically significant change in out-of-pocket premium spending among adults ages 19–26 or those ages 50 and older.

6.4 Robustness Checks

The main analysis in this paper relies on two core assumptions of the standard difference-in-differences approach: that treatment and control groups followed parallel trends before 2019, and that individuals did not change their behavior in anticipation of the mandate repeal. To address potential concerns about these assumptions, I conducted a series of robustness checks, including alternative model specifications, sensitivity analyses, placebo tests, and falsification tests. These exercises help to confirm that the main findings are consistent and reliable.

In this section, I begin with Appendix Figure B.1, which plots trends for newly uninsured, premium spending, and medical out-of-pocket expenditure between the treatment and control states using the CPS-ASEC raw data. Consistent with Figures 4, and 5, the data show an increase in the probability of becoming newly uninsured and declines in out-of-pocket premium spending following the policy change. These results are similar across specifications and do not appear to be driven by pre-existing differences between treatment and control states. The findings provide additional support for the causal interpretation that the observed changes are attributable to the policy.

Next, Appendix Table C.1 reports estimates for two alternative insurance outcomes: any insurance coverage and currently uninsured. Column 1 shows that repealing the ACA individual mandate reduced the probability of having health insurance by 1.56 percentage points, equivalent to approximately a 2 percent decline relative to the pre-reform mean in states without a mandate. Consistent with this result, Column 2 shows that the repeal increased the probability of being uninsured by 1.56 percentage points, or approximately 9 percent relative to the pre-reform mean. These findings closely mirror the baseline estimates and reinforce the conclusion that repealing the individual mandate reduced insurance coverage.

Table C.2 presents the results from the two-way fixed effects model without accounting for mass zeros in out-of-pocket premium spending outcome. I show that the estimates are broadly consistent with the baseline results from the two-part model, though they differ slightly in magnitude due to the presence of mass points at zero. Likewise, the event study results in Appendix Figure C.1 show that the estimates from both the two-part model and the two-way fixed effects approach are nearly identical.

Appendix Table D.1 presents the estimates from two placebo tests that examine the effects of repealing the individual mandate on self-employment and labor force participation.¹⁰ For both placebo outcomes, the estimates are close to zero and statistically insignificant, which further supports that the treatment and control groups did not experience any differential changes after the mandate was repealed. These results provide additional reassurance that my main estimates are not driven by spurious correlations or omitted factors.

Table D.2 of the Appendix reports the falsification test that evaluates the effects of the repeal on two different populations that are not directly affected by the policy. I estimate a set of regressions for elderly and low-income adults, both of whom are less likely to be affected by the repeal of the mandate. Elderly adults aged 65 and older have uninterrupted access to Medicare coverage. In addition, because this population has greater healthcare needs, they are less likely to drop insurance during the study period. On the other hand, low-income non-elderly adults whose household income is below 138% of the federal poverty line (FPL) are eligible for Medicaid and, at the same time, were exempt from the mandate penalty during the entire study period; therefore, they are unlikely to be affected by the repeal of the ACA individual mandate. Consistent with this expectation, the estimated effects for both groups are close to zero. There is no evidence of systematic changes in coverage or spending for these populations around the repeal. These results lend additional support to my causal interpretation that the main estimates capture the policy's effect rather than unrelated trends.

In Appendix Table D.3, I present the sensitivity analysis for different sets of regressions. First,

¹⁰I selected these two outcomes because they are unrelated to the policy and are therefore less likely to be influenced by the policy change.

I estimate the regression model that includes individuals below 138% of the federal poverty line. This specification expands the sample to include low-income individuals, allowing me to assess whether the results are sensitive to changes in the sample composition. Second, given concerns in the prior literature about the treatment group being larger and more diverse than the control group (MA, NJ, and DC), there is a possibility of over-rejecting the null hypothesis. I address this concern following [Soni \(2022\)](#) by restricting my analysis to states within the New England and Mid-Atlantic Census Divisions to ensure a better geographical match between the treatment and control states. I also estimate a regression model that excludes 2018 from my analysis, given that the repeal was announced in 2017 and some individuals might have incorrectly assumed it would take effect in 2018. The estimates across these different specifications and main results are essentially identical to those of the baseline model. While the specifications for the out-of-pocket premium spending and out-of-pocket expenditure are statistically significant at the 1% level, those for the newly uninsured outcome are significant at the 5% level.

Given the decline in insurance coverage following the repeal of the individual mandate, I next examine whether healthcare utilization also changed. Appendix Table [D.4](#) and Figure [D.1](#) report the regression and event-study estimates for routine checkups. I find that the repeal reduced the probability of receiving a routine checkup in the treatment states. This decline is driven entirely by relatively healthy consumers, whose use of preventive care fell following the repeal. By contrast, I find no statistically significant change in routine checkups among relatively sick consumers, consistent with their greater healthcare needs and continued demand for medical care.

Appendix Figure [D.2](#) presents the trends in the United States internet search for "ACA individual mandate" and "Individual Mandate Repeal" from January 2012 to January 2025. The result shows a sharp increase in searches for "Individual Mandate Repeal" in December 2017, around the time the repeal was passed, suggesting strong public interest and awareness of the policy change ([Soni, 2022](#)). Search activity then declines gradually after the initial spike. However, it remains above pre-repeal levels in the subsequent periods. This pattern is consistent with continued

attention to the policy as its effects unfolded.

7 Concluding Remarks

A key justification for the Affordable Care Act (ACA) individual mandate penalty is that it mitigates adverse selection by encouraging both healthy and sick individuals to participate in the health insurance market. At the same time, critics argue that the mandate restricts individual choice and imposes financial burdens on individuals with relatively low demand for insurance. Evaluating these competing perspectives requires understanding not only how the mandate affects insurance participation, but also how its repeal affects coverage and out-of-pocket premium spending across consumers by health status.

To guide my empirical analysis, I build on the workhorse model of [Geruso et al. \(2023a\)](#), which distinguishes between insurance demand at the extensive margin (whether to purchase insurance) and the intensive margin (the generosity of insurance coverage). The model predicts that changes in mandate policy can affect both the decision to purchase insurance and the choice of insurance generosity, with implications for equilibrium premiums and welfare. My analysis applies this framework to the repeal of the ACA individual mandate in order to generate testable predictions about changes in insurance coverage and out of pocket premium payments driven by the repeal.

I test this prediction using data from the CPS ASEC and RWJF health exchange. I find that repealing the ACA individual mandate increased adverse selection in the individual insurance market. The repeal increased the probability of becoming newly uninsured and reduced insurance participation, with these responses driven almost entirely by relatively healthy individuals. In contrast, insurance participation among relatively sick consumers remained largely unchanged, consistent with their stronger demand for health insurance. I further show that although Marketplace premiums increased following the repeal, average out-of-pocket premium spending declined, particularly among healthier consumers. Together, these findings demonstrate that in-

creases in equilibrium premiums do not necessarily translate into higher premium payments for consumers.

I then combine the reduced-form estimates with a simple welfare framework to examine how mandate repeal affected consumer surplus. The analysis shows that, in the absence of premium subsidies, the increase in adverse selection primarily increases the financial burden bear by relatively sick consumers who remain insured because they face higher equilibrium premiums. However, accounting for ACA premium subsidies substantially changes the distributional consequences of the mandate repeal. The ACA Premium subsidies shield many insured consumers from the increase in gross Marketplace premiums, particularly those with greater healthcare needs, thereby mitigating the financial burden generated by adverse selection. At the same time, healthier individuals who leave the insurance market experience modest short-run welfare gains through lower premium expenditures despite forgoing insurance coverage.

More broadly, the results have important implications for the design of health insurance policy beyond the ACA. Because adverse selection following mandate repeal is driven primarily by healthier individuals at the margin of insurance participation, policies aimed at stabilizing insurance markets should be evaluated jointly with the subsidy mechanisms that determine consumers' effective premium payments. Ignoring the interaction between adverse selection and premium subsidies may substantially overstate the financial burden borne by insured consumers and thus may lead to misleading conclusions about the distributional effects of selection when health insurance enrollment becomes voluntary. Future research should examine the longer-run consequences of mandate repeal for insurer participation, market competition, and consumer welfare as insurance markets continue to adjust over time.

Table 1: Descriptive Statistics

	Control States		Treated States		DiD [(4) - (3)] - [(2) - (1)]
	Pre-2019 (1)	2019 (2)	Pre-2019 (3)	2019 (4)	(5)
Insurance Outcomes					
Any Coverage Now (0/1)	0.871	0.906 ^b	0.832 ⁺⁺⁺	0.849 ^{***^^^}	-0.018 ^{‡‡‡}
Newly Insured (0/1)	0.017	0.009 ^b	0.023 ⁺⁺⁺	0.0075 ^{***}	-0.008 ^{‡‡‡}
Newly Uninsured (0/1)	0.014	0.012	0.021 ⁺⁺⁺	0.025 ^{***^^^}	0.006 ^{‡‡‡}
Out-of-Pocket Premium Spending (2015 U.S. \$)	1,740.83	2,044.19 ^a	1,610.69 ⁺⁺⁺	1,601.07 ^{***}	-312.98 ^{‡‡‡}
Marketplace Premium Outcomes					
27yrs old Individual (2015 U.S. \$)	4,485.72	5,107.66 ^a	3,717.30 ⁺⁺⁺	5,402.78 ^{***}	1,063.54 ^{‡‡‡}
50yrs old Individual (2015 U.S. \$)	6,483.24	7,539.95 ^a	6,097.12 ⁺⁺⁺	8,712.71 ^{***}	1,558.88 ^{‡‡‡}
30yrs old Couple (2015 U.S. \$)	10,286.73	11,927.73 ^a	8,523.65 ⁺⁺⁺	13,128.78 ^{***}	2,964.13 ^{‡‡‡}
Individual Characteristics					
Age (years)	40.79	40.62	40.15 ⁺⁺	40.21 ^{**^^}	0.23
Male (0/1)	0.48	0.47	0.48 ⁺⁺	0.48 ^{^^^}	0.01
Married (0/1)	0.46	0.43	0.49 ⁺⁺⁺	0.49 ^{^^^}	0.03
Household Size (number of persons)	3.20	3.26 ^b	3.19 ⁺⁺⁺	3.21 ^{^^^}	-0.04
White non-Hispanic (0/1)	0.52	0.50	0.55 ⁺⁺⁺	0.53 ^{***^^^}	0.00
Black, non-Hispanic (0/1)	0.16	0.18	0.15 ⁺⁺⁺	0.15 ^{**^^^}	-0.02
Other race, non-Hispanic (0/1)	0.09	0.10	0.08	0.08	-0.01
Hispanic (0/1)	0.23	0.22	0.23 ⁺⁺⁺	0.24 ^{***^^^}	0.02
Less Than High School (0/1)	0.10	0.09	0.12 ⁺⁺⁺	0.12 ^{^^^}	0.01
High School (0/1)	0.35	0.39 ^b	0.35	0.35	-0.04
Some College (0/1)	0.30	0.25 ^b	0.33 ⁺⁺⁺	0.32 ^{***^^^}	0.04
College and Advance Degrees (0/1)	0.25	0.27	0.20 ⁺⁺⁺	0.21 ^{**^^^}	-0.01
Working but not Self Employed (0/1)	0.69	0.68	0.69	0.69	0.01
Self-Employed (0/1)	0.06	0.05	0.06 ⁺⁺⁺	0.06 ^{^^^}	0.01
Not working (0/1)	0.25	0.27 ^c	0.25	0.25	-0.02
Sample Size	6,591	1,456	153,965	34,750	

Notes: Sample includes adults aged 19 to 64 years old with household income between 138% and 400% of the federal poverty line (N=196,762). Table displays means adjusted by CPS ASEC sampling weights. Unless otherwise noted, all variables are constructed from the 2015–2019 CPS ASEC. Data on Marketplace premiums are obtained from the 2015–2019 Robert Wood Johnson Foundation (RWJF) Health Insurance Exchange dataset.

Test of significance (Sample means)

Control States, pre- vs post-repeal difference (two-tailed t-test): ^ap<0.01, ^bp<0.05, ^cp<0.1

Pre-repeal treated vs. control states difference (two-tailed t-test): ⁺⁺⁺p<0.01, ⁺⁺p<0.05, ⁺p<0.1

Treated States, pre- vs post-repeal difference (two-tailed t-test): ^{***}p<0.01, ^{**}p<0.05, ^{*}p<0.1

Treated vs Control States difference (two-tailed t-test): ^{^^^}p<0.01, ^{^^}p<0.05, [^]p<0.1

Test of significance (DiD estimates): ^{‡‡‡}p<0.01, ^{‡‡}p<0.05, [‡]p<0.1.

Table 2: Regression Results for Insurance Outcomes (Source: CPS ASEC, 2015 to 2019)

	Newly Uninsured			Newly Insured		
	All (1)	Healthy (2)	Sick (3)	All (4)	Healthy (5)	Sick (6)
Treatment X Post2019	0.0056*** (0.0018)	0.0057*** (0.0014)	0.0039 (0.0075)	-0.0079*** (0.010)	-0.0087*** (0.0015)	-0.0020 (0.0071)
Sample Size	196,762	172,899	23,863	196,762	172,899	23,863
Pre-2019 Mean	0.0208	0.0213	0.0167	0.023	0.022	0.028
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes	Yes
Individual Controls	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Sample includes adults aged 19 to 64 years old with household income between 138% and 400% of the federal poverty line (N=196,762). Table displays coefficient estimate and state clustered standard error for the interaction of an indicator for residence in a state that repealed the insurance mandate and the probability of becoming newly uninsured for full, healthy and sick samples. All regressions control for age, sex, marital status, household size, race/ethnicity, educational attainment, employment status, state fixed effects, and year effects. Estimates are weighted by CPS ASEC sampling weights. ***p<0.01, **p<0.05, *p<0.10. Standard errors are in parentheses.

Table 3: Regression Estimates for Out-of-Pocket Premium Spending (Source: CPS ASEC, 2015 to 2019)

	All (1)	Healthy (2)	Sick (3)
Treatment X Post2019	-284.60*** (35.93)	-282.59*** (43.05)	-233.74 (200.43)
Sample Size	196,762	172,899	23,863
Pre-2019 Mean	1,615.97	1,660.90	1,299.28
Year FE	Yes	Yes	Yes
State FE	Yes	Yes	Yes
Individuals Controls	Yes	Yes	Yes

Notes: Sample includes adults aged 19 to 64 years old with household income between 138% and 400% of the federal poverty line (N=196,762). Table displays the coefficient estimate and state clustered standard error for the interaction of an indicator for residence in states that repealed the insurance mandate on medical out-of-pocket expenditure for both the extensive and intensive margins as well as the combined effects. All regressions control for age, sex, marital status, household size, race/ethnicity, educational attainment, employment status, state fixed effects, and year effects. Estimates are weighted by CPS ASEC sampling weights. ***p<0.01, **p<0.05, *p<0.10. Standard errors are in parentheses.

Table 4: Regression Estimates for Market Premium (Source: RWJF Health Exchange 2015 to 2019)

	27yrs-old (1)	50yrs-old (2)	Couple 30yrs-old (3)
Treatment X Post2019	738.93*** (110.98)	1,361.46*** (210.99)	3,341.60*** (426.12)
Sample Size	293,967	293,967	293,967
Pre-2019 Mean	3,478.95	5,819.43	11,851.59
Year FE	Yes	Yes	Yes
State FE	Yes	Yes	Yes
Individual Controls	No	No	No

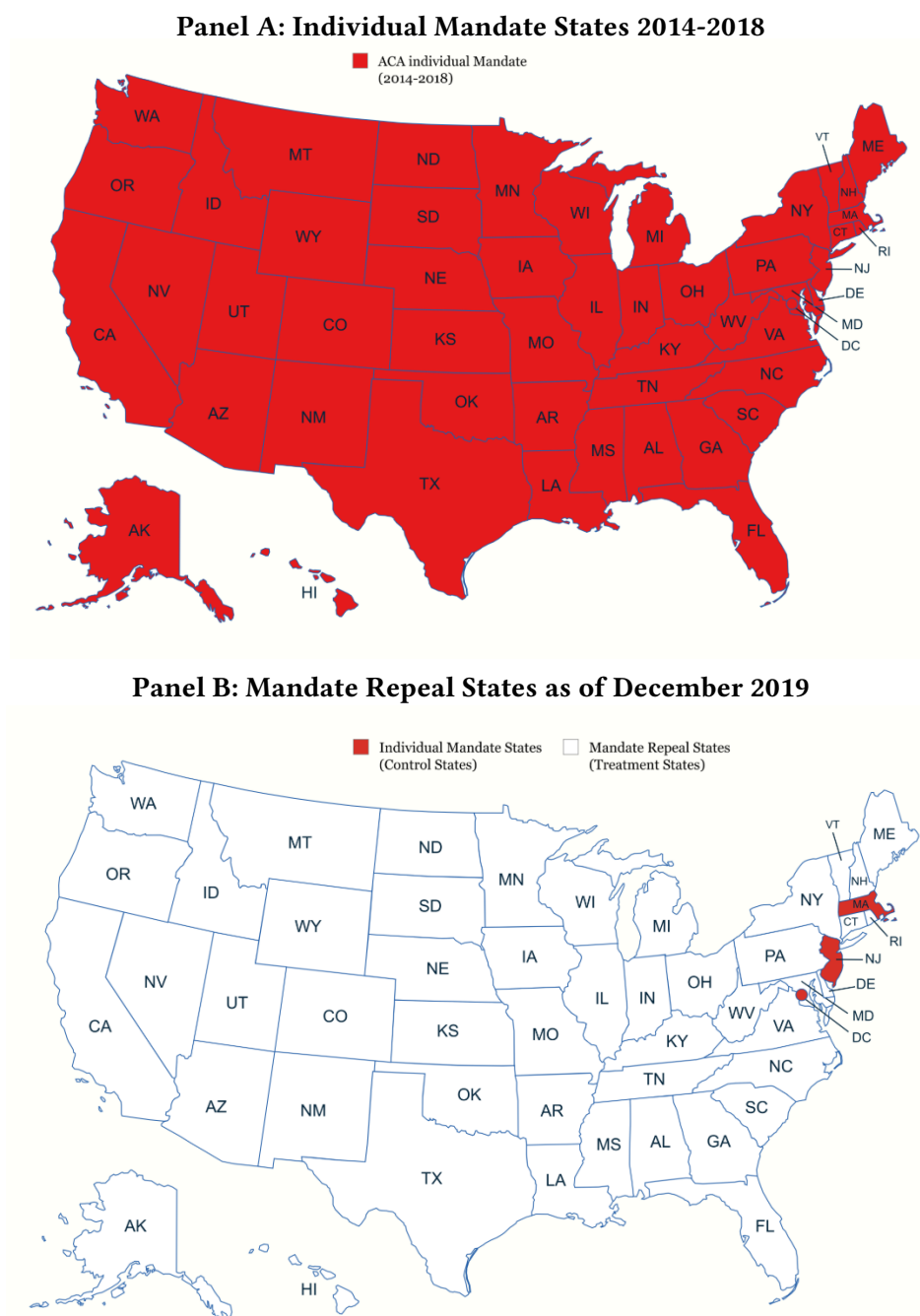
Notes: Sample includes insurance plans across U.S. health exchange markets (N=293,967). Table displays coefficient estimate and state clustered standard error for the interaction of an indicator for residence in a state that repealed the insurance mandate and the market premium price for individuals and couples. All regressions control for state fixed effects, and year effects. The insurance outcomes are weighted using the CPS ASEC sampling weights. While the market premium price estimates are weighted by state population. ***p<0.01, **p<0.05, *p<0.10. Standard errors are in parentheses.

Table 5: Heterogeneity Analysis for Healthy Consumers (Source: CPS ASEC, 2015 to 2019)

	Newly Uninsured (1)	Newly Insured (2)	Out of Pocket Premium Spending (3)	Sample Size (4)
Full Sample	0.0056*** (0.0018)	0.0079*** (0.0001)	-338.48*** (0.0010)	196,762
Healthy	0.0057 *** (0.004)	-0.0087*** (0.0015)	-282.59*** (43.05)	172,899
By Sex				
Male	0.0059*** (0.0019)	-0.0022 (0.0026)	-177.26*** (53.04)	82,175
Female	0.0054** (0.0023)	-0.0141*** (0.0040)	-357.99*** (58.06)	90,724
By Marital Status				
Married	0.0020 (0.0018)	-0.001 (0.0015)	-453.70*** (65.31)	93,156
Not Married	0.0093*** (0.0023)	-0.0157*** (0.0036)	-157.44*** (44.02)	79,743
By Parental Status				
Parents	0.0041 (0.0030)	-0.0093*** (0.0016)	-287.71*** (90.68)	98,393
Childless Adults	0.0073*** (0.0026)	-0.0084*** (0.0020)	-262.41*** (51.38)	74,506
By Employment Status				
Working but not Self-employed	0.0026 (0.0048)	-0.0007 (0.0029)	-410.12*** (103.33)	46,065
Self-employed	0.0067*** (0.0018)	-0.0116*** (0.0013)	-233.30*** (51.21)	126,834
Not Working	-	-	-	
By Educational Status				
Less than High School	0.0060 (0.0043)	-0.0273*** (0.0081)	-381.93*** (189.55)	19,879
High School	0.0127** (0.0049)	0.0024 (0.0038)	-108.29 (133.36)	58,678
Some College	-0.0078 (0.0118)	-0.0228*** (0.0023)	-415.19*** (60.74)	57,501
College and Advanced Degrees	0.0088 *** (0.0029)	-0.0041 (0.0066)	-360.81*** (74.23)	36,841
By Race				
White	0.0165*** (0.0029)	-0.0108*** (0.0035)	-173.56** (69.65)	92,065
Black	-0.0033 (0.0045)	-0.0096*** (0.0029)	-49.25 (71.94)	22,134
Hispanic	-0.0182*** (0.0030)	-0.0040 (0.0072)	-612.82*** (103.58)	42,905
Other Race	0.0208*** (0.0051)	-0.0104 (0.0140)	-487.36*** (127.27)	15,795
By Age				
19-26	0.0191* (0.0104)	-0.0274** (0.0103)	-101.28 (107.95)	33,113
27-39	-0.0063 (0.0116)	0.0052 (0.0053)	-491.91*** (109.52)	59,190
40-49	0.0009 (0.0041)	-0.0022 (0.0027)	-486.67** (195.89)	38,569
50-59	0.0063** (0.0028)	-0.0022 (0.0029)	-50.69 (93.29)	29,421
60-64	0.0274*** (0.0059)	-0.0318*** (0.0043)	-231.97 (403.62)	12,606

Notes: Sample includes adults aged 19 to 64 years old with household income between 138% and 400% of the federal poverty line (N=196,762). Table displays coefficient estimate and state clustered standard error for the interaction of an indicator for residence in a state that repealed the insurance mandate and consumer premium spending outcome. All regressions control for age, sex, marital status, household size, race/ethnicity, educational attainment, employment status, state fixed effects, and year effects. Estimates are weighted by CPS ASEC sampling weights. ***p<0.01, **p<0.05, *p<0.10. Standard errors are in parentheses.

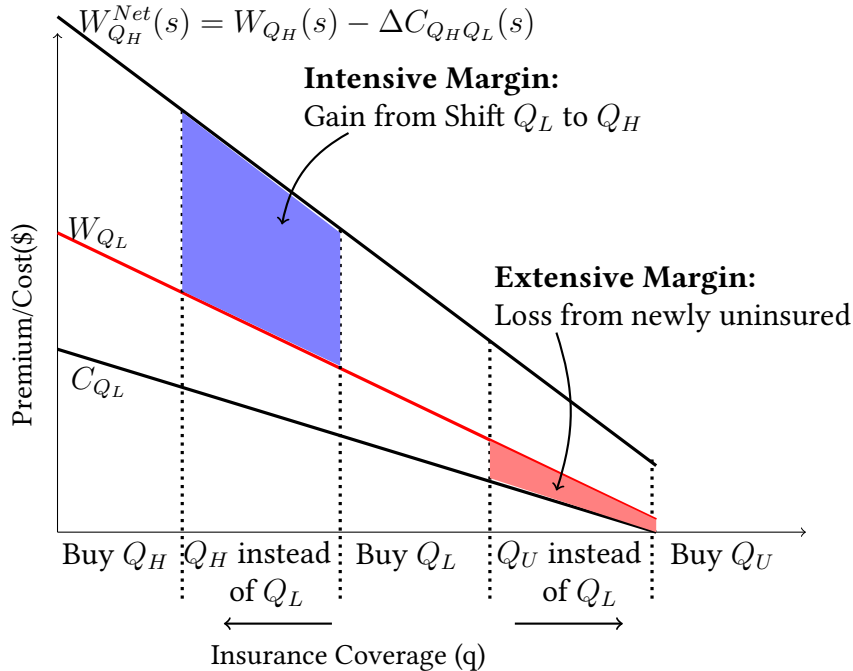
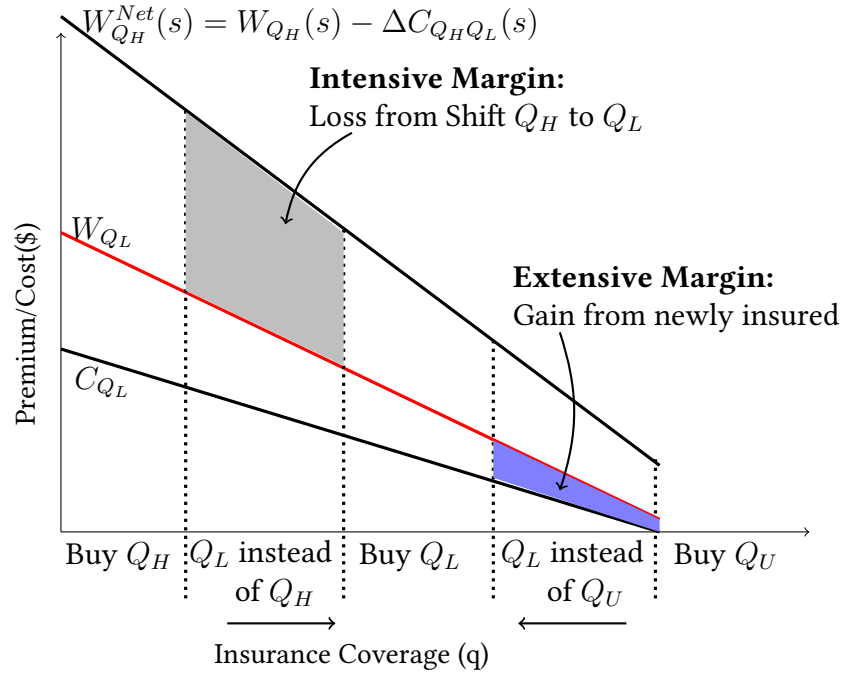
Figure 1: Individual Mandates and Repealed States as of December 2019



Note: Panel A of Figure 1 shows the implementation of the ACA individual mandate across the 50 states and the District of Columbia between 2014 and 2018. Panel B identifies the treatment states (those without an individual mandate in white) and the control states (those retaining an individual mandate in red) following the repeal of the federal mandate penalty. After the federal individual mandate penalty was reduced to zero in January 2019, Massachusetts, New Jersey, and the District of Columbia implemented state-level individual mandates, leaving 48 states without mandates.

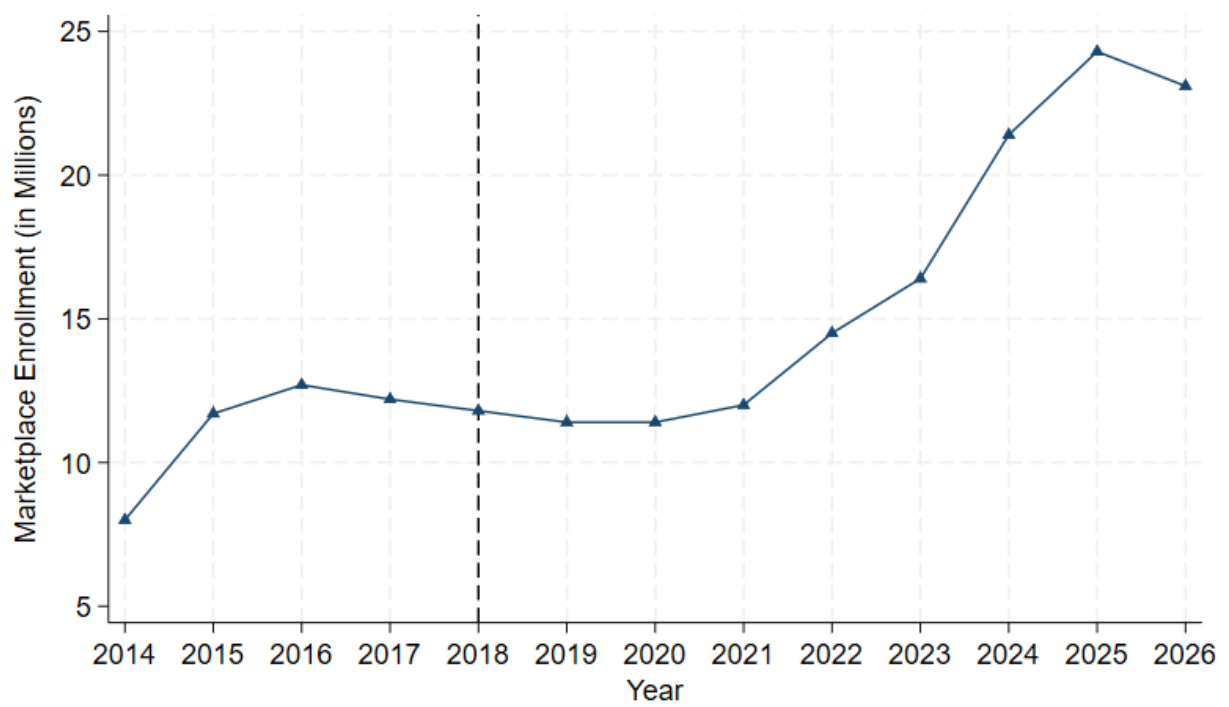
Figure 2: Effects of Adverse Selection Consumer

Panel A: Consumer Surplus Under Mandate Penalty



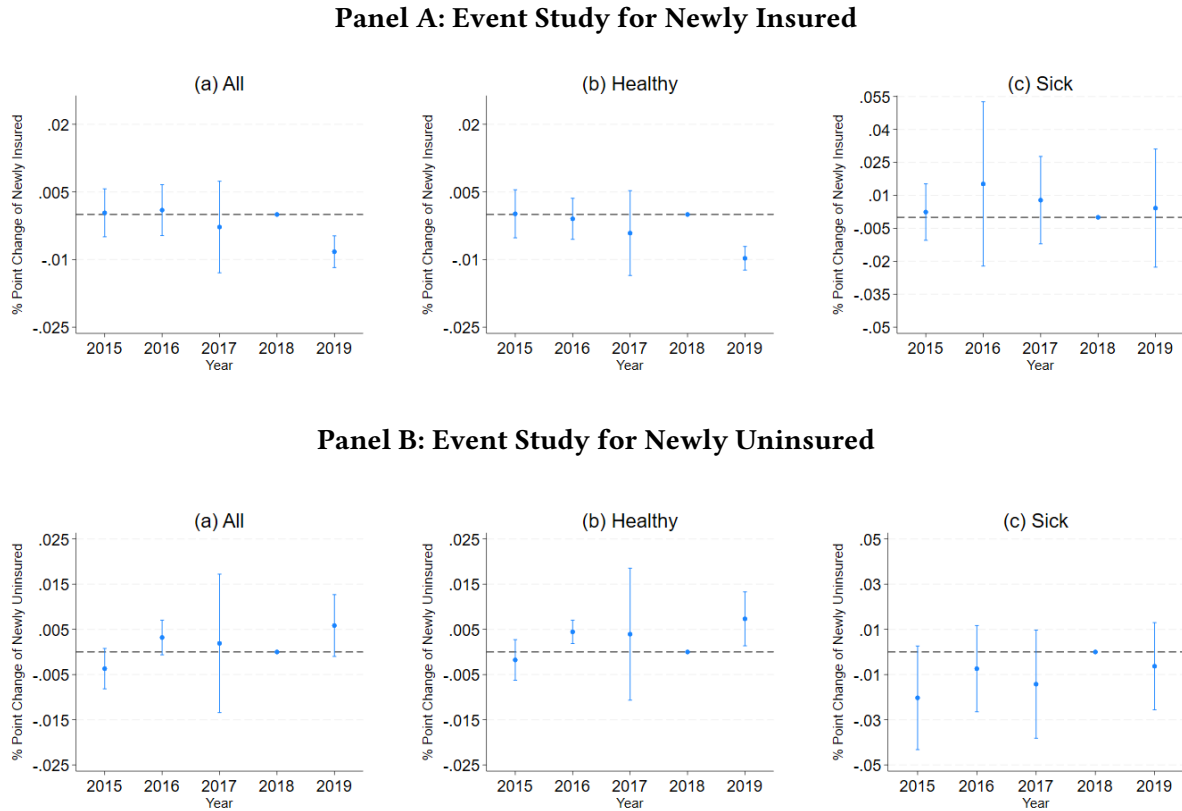
Note: Figure 2 illustrates the demand and cost curves in the health insurance market. Consumers are ordered along the horizontal axis according to their willingness to pay for insurance. The socially efficient allocation occurs at $Q^{*,pre}$, where the demand curve, $D(Q)$, intersects the marginal cost curve, $MC(Q)$. In contrast, the competitive equilibrium is at Q^{pre} , where the demand curve intersects the average cost curve $AC(Q)$ reflecting the presence of adverse selection. The red shaded triangle represents the forgone

Figure 3: Trends in ACA Marketplace Enrollment



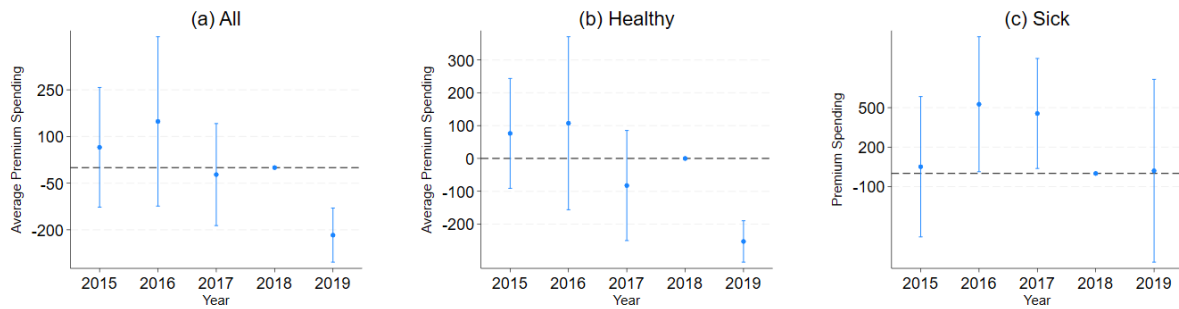
Note: The figure display the trends in the ACA marketplace enrollment across health exchange. (Source: Center for Medicare and Medicaid Services(CMS) Report, 2014 to 2026.

Figure 4: Effects of the repeal of the mandate on Insurance Outcome



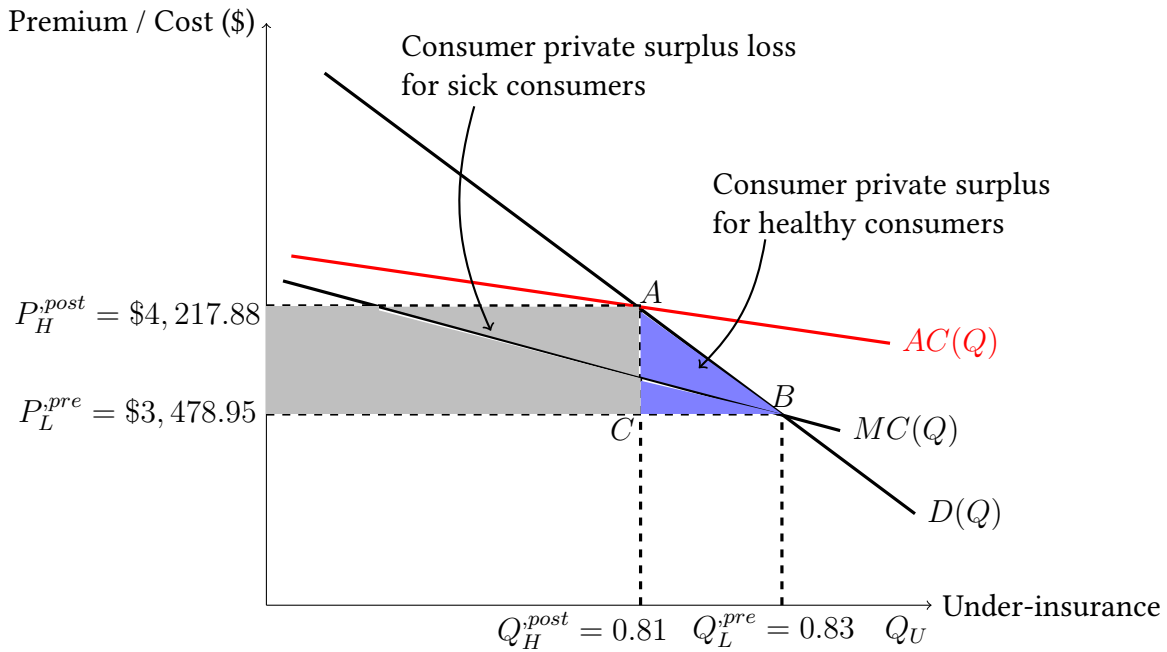
Note: Figure 4 reports the event-study estimates for the probability of becoming newly insured, while Panel B reports the corresponding estimates for the probability of becoming newly uninsured. Estimates are shown separately for healthy consumers, sick consumers, and the full sample, comparing repeal states with states that retained an individual mandate. The sample includes adults ages 19–64 with household incomes between 138% and 400% of the federal poverty level ($N = 196,762$). Estimates are normalized to zero in 2018, the year immediately preceding the repeal. Vertical bars represent 95% confidence intervals. All regressions control for age, sex, marital status, household size, race/ethnicity, educational attainment, employment status, state fixed effects, and year fixed effects. Standard errors are clustered at the state level, and all estimates are weighted using CPS ASEC sampling weights.

Figure 5: Effects of the Repeal of the ACA Mandate on Out-of-Pocket Premium Spending



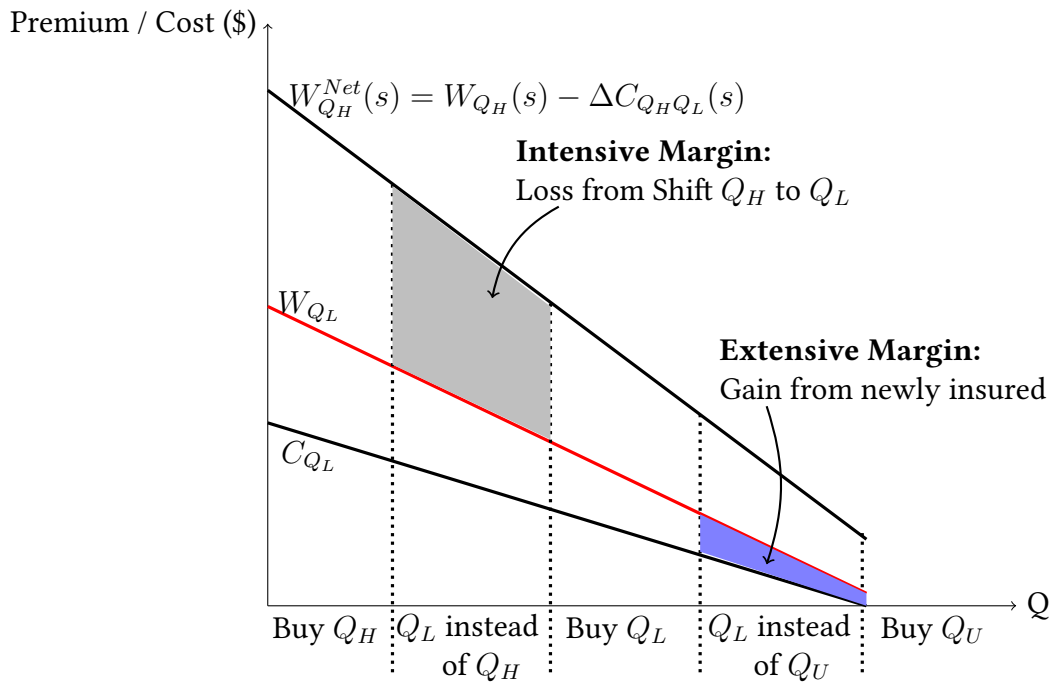
Note: Figure 5 presents the event-study estimates of the effects of repealing the ACA individual mandate on out-of-pocket premium spending. The figure reports event-study estimates for the full sample and separately for healthy and sick consumers, comparing states without state-level mandates to states that retained a mandate. Estimates are normalized to zero in 2018, the year immediately preceding the repeal. The sample includes adults ages 19–64 with household incomes between 138% and 400% of the federal poverty level (N=196,762). Points represent coefficient estimates, and vertical bars denote 95% confidence intervals. All regressions control for age, sex, marital status, household size, race/ethnicity, educational attainment, employment status, state fixed effects, and year fixed effects. Standard errors are clustered at the state level, and all estimates are weighted using CPS ASEC sampling weights. Source: Current Population Survey Annual Social and Economic Supplement (CPS ASEC), 2015–2019.

Figure 6: Consumer Surplus Under Adverse Selection



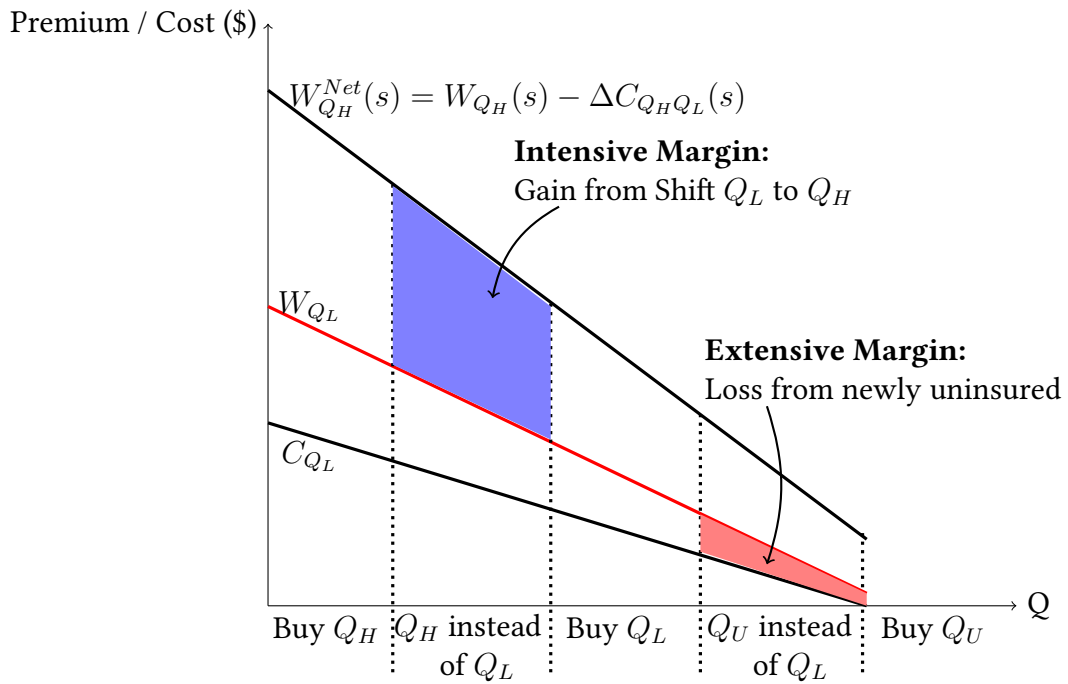
Note: Figure 6 illustrates the demand and cost curves in the health insurance market. Consumers are ordered along the horizontal axis according to their willingness to pay for insurance. The socially efficient allocation occurs at $Q_L^{*,pre} = 0.83$, where the demand curve, $D(Q)$, intersects the marginal cost curve, $MC(Q)$. In contrast, the competitive equilibrium is at $Q_H^{post} = 0.81$, where the demand curve intersects the average cost curve, $AC(Q)$, reflecting the presence of adverse selection. The blue shaded triangle represents the forgone consumer surplus of relatively healthy (marginal) consumers who do not purchase insurance, while the gray shaded rectangle captures the transfer in consumer surplus borne by relatively sick (inframarginal) consumers who remain insured and pay higher premiums.

Figure 7: Consumer Surplus Under Adverse Selection



Note: Figure 2 illustrates the demand and cost curves in the health insurance market. Consumers are ordered along the horizontal axis according to their willingness to pay for insurance. The socially efficient allocation occurs at $Q^{*,pre}$, where the demand curve, $D(Q)$, intersects the marginal cost curve, $MC(Q)$. In contrast, the competitive equilibrium is at Q^{pre} , where the demand curve intersects the average cost curve, $AC(Q)$, reflecting the presence of adverse selection. The red shaded triangle represents the forgone consumer surplus of relatively healthy (marginal) consumers who do not purchase insurance, while the blue shaded rectangle captures the transfer in consumer surplus borne by relatively sick (inframarginal) consumers who remain insured and pay higher premiums.

Figure 8: Consumer Surplus Under Adverse Selection



Note: Figure 2 illustrates the demand and cost curves in the health insurance market. Consumers are ordered along the horizontal axis according to their willingness to pay for insurance. The socially efficient allocation occurs at $Q^{*,pre}$, where the demand curve, $D(Q)$, intersects the marginal cost curve, $MC(Q)$. In contrast, the competitive equilibrium is at Q^{pre} , where the demand curve intersects the average cost curve, $AC(Q)$, reflecting the presence of adverse selection. The red shaded triangle represents the forgone consumer surplus of relatively healthy (marginal) consumers who do not purchase insurance, while the blue shaded rectangle captures the transfer in consumer surplus borne by relatively sick (inframarginal) consumers who remain insured and pay higher premiums.

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Appendix

A Theory of Adverse Selection and Individual Mandate

In this section, I build a simple model by incorporating the features of the ACA individual mandate and premium subsidy into a general model of adverse selection built by [Rothschild and Stiglitz \(1976\)](#). This model captures consumers' behavior by risk type when the insurance mandate is turned on or off at the extensive margin. This modeling framework is motivated by the ACA's individual mandate and premium subsidy features, which primarily influence the demand for health insurance on the extensive margin.

A.1 Set Up

In health insurance contracts, there are three participants in the market: individuals who purchase insurance, healthcare management organizations that sell it, and hospitals/doctors who provide healthcare. In this paper, I focus on consumer demand for health insurance. I first consider a case of a competitive consumer with endowment Y . If he does not fall sick, then his endowment Y remains the same. However, if he falls sick, then the consumer incurs a loss $L < Y$, so that his endowment will be $Y-L$. Without health insurance, his endowment in the two states, "healthy" and "sick", is $(Y, Y-L)$.

There is also a competitive market that offers different health insurance plans for consumers to buy. The consumer can buy health insurance coverage q to protect himself against illness by paying a premium p per unit of compensation in U.S. dollars in the event he falls ill. In return, he receives insurance coverage q (equivalent to $\alpha \in [0, 1]$ of the total medical costs) in the event he falls sick. With health insurance, his endowment in the two states "healthy" and "sick" is $(Y-p, Y-p-(1-\alpha)L)$. The expected utility of the competitive consumer is:

$$EU[\pi_i, 0, 0, 0] = \underbrace{U(Y)(1 - \pi_i)}_{\text{Healthy State}} + \underbrace{U(Y - L)(\pi_i)}_{\text{Sick State}}$$

Where $U()$ denotes the utility of his endowment, and π_i is the probability of falling sick and $i \in \{Sick, Healthy\}$ consumer type. I assume the consumer's utility function is strictly concave and all individuals (both the sick and healthy) are risk-averse ($U'' < 0$).

There is also a competitive market that offers different health insurance plans for consumers to buy.

The consumer can buy health insurance coverage q to protect himself against illness by paying a premium p per unit of compensation in U.S. dollars in the event he falls ill. In return, he receives insurance coverage q (equivalent to $\alpha \in [0, 1]$ of the total medical costs) in the event he falls sick. With health insurance, his endowment in the two states "healthy" and "sick" is $(Y-p, Y-p-(1-\alpha)L)$. Therefore, his expected utility is:

$$EU[\pi_i, q, 0, 0] = \underbrace{U(Y-p)(1-\pi_i)}_{\text{Healthy State}} + \underbrace{U[(Y-p-(1-\alpha)L](\pi_i)}_{\text{Sick State}}$$

Without an insurance mandate, he will always have the option of purchasing no health insurance. In this case, a consumer will purchase a health insurance contract q only if $EU[\pi_i, q, 0, 0] > EU[\pi_i, 0, 0, 0]$. However, a low-risk (healthy) individual may choose not to buy insurance if the actuarially fair premium is too high relative to their expected healthcare costs.

A.1.1 Government Intervention through Individual Mandate and Premium Subsidy

This section considers a case in which the government intervenes through a mandate penalty and premium subsidy. Governments sometimes intervene through policy tools such as premium subsidies and individual mandates to mitigate the risk of market unraveling due to selection. While premium subsidy interventions provide an incentive for consumers, particularly lower-income and low-risk (healthy) enrollees, to buy insurance, who might otherwise remain uninsured, individual mandates compel them to purchase insurance or face a tax penalty. I modified the RS model to include key features of the ACA (premium subsidy and individual mandate tax penalty) and to illustrate consumer behavior by type in the U.S. health insurance market.

Under government intervention through individual mandate and premium subsidy, a consumer may either purchase insurance coverage q to protect himself against illness by paying an actuarially fair premium $p = \pi L$ and receive a subsidy s or remain uninsured and incur a tax penalty τ . Suppose the consumer purchases the insurance to protect against illness, the insurance company will receive the premium and pay an amount equivalent to L when he is sick. Suppose the utility for the consumer when insured is $U[\pi_i, q, s, 0]$ and $U(\pi_i, 0, 0, \tau)$ when uninsured. I further assume that the consumer's utility U is increasing and concave: $U' > 0$ and $U'' < 0$. The expected utility for the healthy individual is given by:

$$\begin{aligned}
EU[\pi_i, q, s, 0] &= \underbrace{U[Y - (p - s)](1 - \pi_i)}_{\text{Healthy State}} + \underbrace{U[Y - (p - s) - (1 - \alpha)L](\pi_i)}_{\text{Sick State}} \\
EU[\pi_i, 0, 0, \tau] &= \underbrace{U(Y - \tau)(1 - \pi_i)}_{\text{Healthy State}} + \underbrace{U(Y - \tau - L)(\pi_i)}_{\text{Sick State}}
\end{aligned}$$

Where $i \in \{Sick, Healthy\}$. Since I assume consumers are risk-averse, they will always choose to buy health insurance when it is mandatory rather than remain uninsured and face a tax penalty and uncertain future health risk.

A.1.2 Government Intervention through Mandate Repeal and Premium Subsidy

In this section, I consider a setting in which the government repeals the individual mandate but continues to provide premium subsidy. Under this setting, consumers have the choice to become uninsured and face a zero tax penalty or continue to buy insurance coverage to protect against illness by paying an actuarially fair premium $p = \pi L$ and receive a subsidy s from the government. Suppose the consumer purchases the insurance to protect against illness, the insurance company will receive the premium and pay an amount equivalent to L when he is sick. I also assume that consumers fall into two categories: healthy and sick. Healthy consumers face relatively low expected medical costs, with a true risk $\pi_H \approx 0$. On the other hand, the sick consumers face a higher probability of incurring medical expenditures with true risk $\pi_S > 0$.

Healthy Enrollees: To illustrate how the repeal of the ACA mandate penalty affects insurance market participation, I begin with the healthy consumer type. Healthy consumers face relatively low expected medical costs, with a true risk $\pi_H \approx 0$. Suppose the utility for the healthy consumer when insured is $U_H[\pi_H, q, s, 0]$ and $U_H(\pi_H, 0, 0, 0)$ when uninsured. I further assume that the consumer's utility U is increasing and concave: $U' > 0$ and $U'' < 0$. The expected utility for the healthy individual is given by:

$$\begin{aligned}
EU_H[\pi_i, q, s, 0] &= \underbrace{U[Y - (p - s)](1 - \pi_H)}_{\text{Healthy State}} + \underbrace{U[Y - (p - s) - (1 - \alpha)L](\pi_H)}_{\text{Sick State}} \\
EU_H[\pi_i, 0, 0, 0] &= \underbrace{U(Y)(1 - \pi_H)}_{\text{Healthy State}} + \underbrace{U[(Y - L)](\pi_H)}_{\text{Sick State}}
\end{aligned}$$

Since healthy consumers face a very low probability of illness, $EU_H[\pi_i, 0, 0, 0] > EU_H[\pi_i, q, s, 0]$. In

the short run, removing the mandate penalty therefore increases the incentive for healthier individuals to remain uninsured and avoid premium costs.

Sick Enrollees: Finally, I consider a case where consumers are of a sick type. Unlike healthy consumers, sick enrollees face a higher probability of incurring medical expenditures, with true risk $\pi_S > 0$. Assume the utility for the sick consumer when insured is $U_S[\pi_S, q, s, 0]$ and $U_S(\pi_S, 0, 0, 0)$ when uninsured. Suppose that the consumer's utility U is increasing and concave: $U' > 0$ and $U'' < 0$. The expected utility for this type of consumer is given by:

$$\begin{aligned}
 EU_S[\pi_i, q, s, 0] &= \underbrace{U[Y - (p - s)](1 - \pi_S)}_{\text{Healthy State}} + \underbrace{U[Y - (p - s) - (1 - \alpha)L](\pi_S)}_{\text{Sick State}} \\
 EU_S[\pi_i, 0, 0, 0] &= \underbrace{U(Y)(1 - \pi_S)}_{\text{Healthy State}} + \underbrace{U[(Y - L)(\pi_S)]}_{\text{Sick State}}
 \end{aligned}$$

The sick enrollees are more likely to be ill. Because $EU_S[\pi_i, q, s, 0] > EU_S[\pi_i, 0, 0, 0]$, they have a stronger incentive to maintain insurance coverage even after the repeal of the mandate penalty.

A.2 Distributional Incidence of Repealing the ACA Individual Mandate

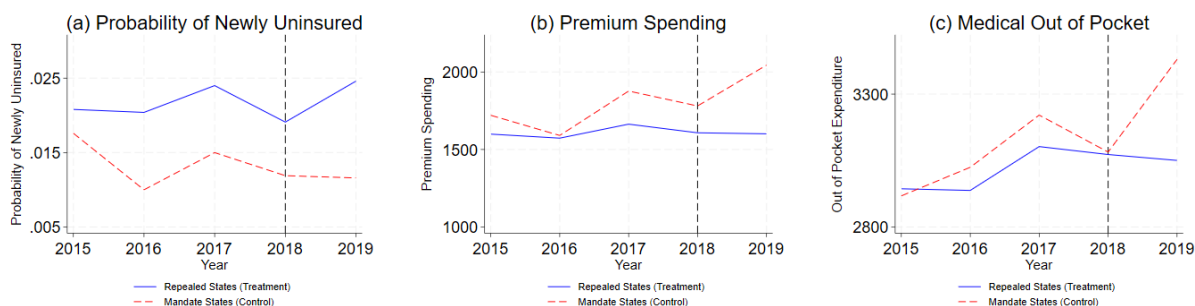
B Individual Mandates and Repeal States

Table B.1: States with Insurance Mandate and States without Mandate as of 2024

	2019	2020
Treatment	48 States	46 states
Control	MA, NJ, DC	MA, NJ, DC, CA, RI

Notes: The table displays treatment states (without mandates) and the control states (with mandates). Following the repeal of the ACA federal individual mandate in January 2019, two states (MA, NJ) and the D.C. implemented the state level mandate leaving 48 states without individual mandate. By January 2020 CA and RI also implemented state-level mandates while the rest 46 states continued without mandate.

Figure B.1: Trends in Newly Uninsured and Healthcare Spending Outcomes



Note: Panel (a) display the trends in the probability of becoming newly uninsured between the treatment and the control states. Panel (b) Display the trend in premium spending by individuals at the time of interview between the treatment and control states. Panel (c) Display trends in medical out-of-pocket spending by individuals at the time of interview in states with and without an insurance mandate. (Source: Current Population Survey ASEC, 2015 to 201.; Notes: Sample includes adults aged 19 to 64 with household income between 138% and 400% of the poverty level (N=196,672). Estimates are weighted by CPS ASEC sampling weights).

C Main Results

Table C.1: Alternate Insurance Outcomes (Source: Current Population Survey ASEC, 2015 to 2019)

	Any Coverage Now (1)	Currently Uninsured (2)
Treatment X Post2019	-0.0156*** (0.0046)	0.0156*** (0.0046)
Sample Size	196,762	196,762
Pre 2019 Mean	0.83	0.17
Year FE	Yes	Yes
State FE	Yes	Yes
Individuals Controls	Yes	Yes

Notes: The sample includes adults aged 19 to 64 whose household income falls between 138% and 400% of the federal poverty line (N=196,762). The table shows the estimated coefficients and state-clustered standard errors for the interaction between living in a state that repealed the insurance mandate and alternate insurance outcomes. All models control for individual characteristics such as age, sex, marital status, household size, race/ethnicity, education level, and employment status, as well as state and year fixed effects. Estimates are weighted by CPS ASEC sampling weights. ***p<0.01, **p<0.05, *p<0.10. Standard errors are in parentheses.

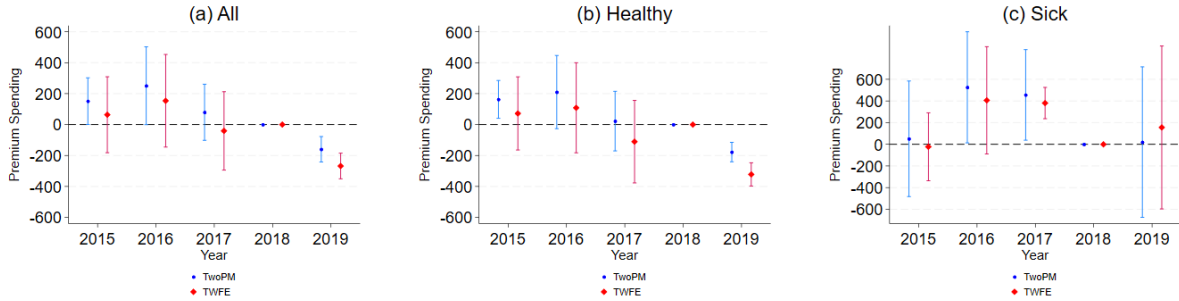
Table C.2: Regression Estimates for Out-of-Pocket Premium Spending, and Medical Out-of-Pocket Costs (Source: CPS ASEC, 2015 to 2019)

	Full Sample TWFE						Insured Only TWFE					
	Premium Spending			Out of Pocket			Premium Spending			Out of Pocket		
	All (1)	Healthy (2)	Sick (3)	All (4)	Healthy (5)	Sick (6)	All (7)	Healthy (8)	Sick (9)	All (10)	Healthy (11)	Sick (12)
Treatment X Post2019	-326.05*** (33.65)	-338.36*** (41.56)	-176.32 (243.38)	-346.99*** (49.61)	-396.09*** (60.73)	63.24 (528.34)	-319.20*** (33.62)	-335.63*** (53.38)	-120.74 (244.97)	-337.96*** (46.63)	-385.39*** (78.27)	70.87 (531.30)
Sample Size	196,762	172,899	23,863	196,762	172,762	23,863	165,529	144,790	20,739	165,529	144,790	20,739
Pre-2019 Mean	1,615.97	1,660.90	1,299.28	3,014.73	2,997.81	3,133.98	1,831.48	1,896.06	1,398.50	3,303.40	3,192.36	3,283.95
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Individuals Control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

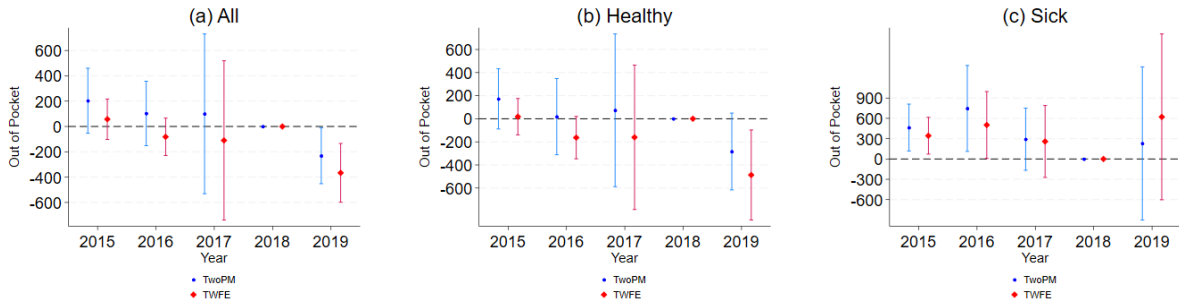
Notes: Sample includes adults aged 19 to 64 years old with household income between 138% and 400% of the federal poverty line (N=196,762). Table displays the coefficient estimate and state clustered standard error for the interaction of an indicator for residence in a state that repealed the insurance mandate alongside premium spending, and out-of-pocket expenditure in both the extensive and intensive margin. All regressions control for age, sex, marital status, household size, race/ethnicity, educational attainment, employment status, state fixed effects, and year effects. Estimates are weighted by CPS ASEC sampling weights. ***p<0.01, **p<0.05, *p<0.10. Standard errors are in parentheses.

Figure C.1: Event Study for Out-of-Pocket Premium Spending

Panel A: Full Sample

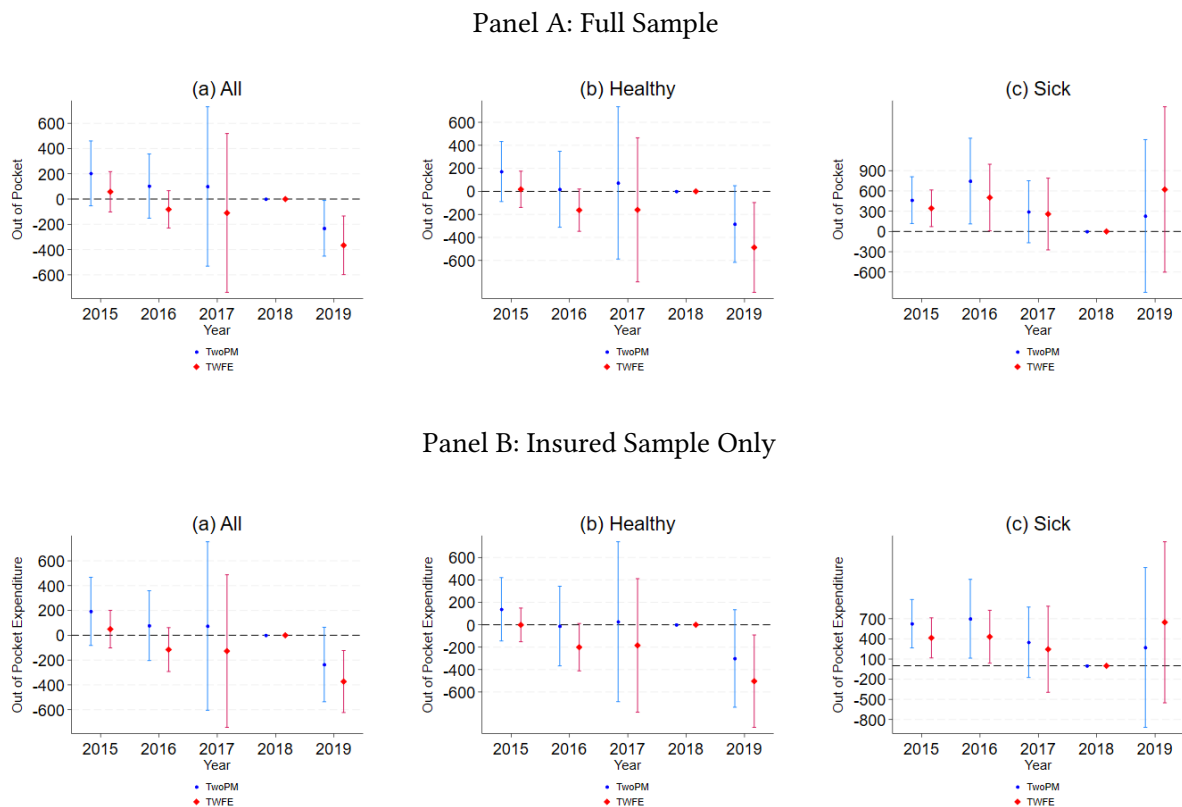


Panel B: Insured Sample Only



Note: Fig B.1 display the differential change in Premium expenditure for individuals with any health insurance coverage in the repeal states relative to states with mandates. Notes: Sample includes adults aged 19 to 64 with household income between 138% and 400% of the poverty level (N=196,762). The Figure displays coefficient estimates and 95% confidence intervals for a vector of variables interacting the mandate repeal indicator with indicator variables for each year; the 2018 indicator is omitted as the reference period. All regressions also control for age, sex, marital status, household size, race/ethnicity, educational attainment, employment status, state fixed effects, and year effects. Standard errors are clustered by state, and estimates are weighted by CPS ASEC sampling weights).

Figure C.2: Event Study for Medical Out of Pocket Expenditure



Note: Fig. B.2 shows the differential change in medical out of pocket expenditure for individuals with any health insurance coverage in the repeal states relative to states with mandates. Notes: Sample includes adults aged 19 to 64 with household income between 138% and 400% of the poverty level (N=189,850). The Figure displays coefficient estimates and 95% confidence intervals for a vector of variables interacting the mandate repeal indicator with indicator variables for each year; the 2018 indicator is omitted as the reference period. All regressions also control for age, sex, marital status, household size, race/ethnicity, educational attainment, employment status, state fixed effects, and year effects. Standard errors are clustered by state, and estimates are weighted by CPS ASEC sampling weights.

D Robustness Checks

Table D.1: Placebo Test for Alternate Outcomes (Source: CPS ASEC, 2015 to 2019)

	Self-employed (1)	In Labor Force (2)
Treatment X Post2019	3.36×10^{-17} (2.28×10^{-17})	0.0005 (0.0012)
Sample Size	149,195	195,250
Controls	Yes	Yes
Year FE	Yes	Yes
State FE	Yes	Yes

Notes: Unless otherwise noted, the sample includes adults ages 19 to 64 with household incomes between 138% and 400% of the federal poverty level (N=196,762). The table reports the coefficient estimate and the state-clustered standard error for the interaction term capturing individuals living in states that repealed the insurance mandate in the year 2019. All regressions also control for age, sex, marital status, household size, race/ethnicity, educational attainment, employment status, state and year fixed effects. ***p<0.01, **p<0.05, *p<0.10. Standard errors are in parentheses.

Table D.2: Falsification Test (Source: Current Population Survey ASEC, 2015 to 2019)

	Newly Uninsured		Premium Spending		Out of Pocket Expenditure	
	Elderly (Age >64) (1)	Low-Income (<138% FPL) (2)	Elderly (Age >64) (3)	Low-Income (<138% FPL) (4)	Elderly (Age >64) (5)	Low-Income (<138% FPL) (6)
Treatment X Post2019	-0.0012*** (0.0003)	0.0054 (0.0046)	56.46 (74.14)	-87.89 (158.01)	-103.90 (66.00)	-29.39 (215.59)
Sample Size	54,962	76,937	54,962	76,937	54,962	76,937
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes	Yes
Individuals Control	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The sample includes adults aged 19 to 64 whose household income falls between 138% and 400% of the federal poverty line (N=196,762). The table shows the estimated coefficients and state-clustered standard errors for the interaction between living in a state that repealed the insurance mandate and consumer premium spending. All models control for individual characteristics such as age, sex, marital status, household size, race/ethnicity, education level, and employment status, as well as state and year fixed effects. Estimates are weighted by CPS ASEC sampling weights. ***p<0.01, **p<0.05, *p<0.10. Standard errors are in parentheses.

Table D.3: Sensitivity Analyses (Source: Current Population Survey ASEC, 2015 to 2019)

	Newly Uninsured				Premium Spending				Medical Out of Pocket Expenditure			
	Baseline	Include Income	New England	Omit 2018	Baseline	Include Income	New England	Omit 2018	Baseline	Include Income	New England	Omit 2018
	0-138%		& Mid-Atlantic		0-138%		& Mid-Atlantic		0-138%		& Mid-Atlantic	
	FPL Sample		Divisions Only		FPL Sample		Divisions Only		FPL Sample		Divisions Only	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Treatment	0.0056***	0.0053**	0.0055**	0.0056***	-284.60***	-276.21***	-293.81***	-323.94***	-338.50***	-319.14***	-301.001*	-373.79***
X Post2019	(0.0018)	(0.0023)	(0.0018)	(0.0017)	(35.93)	(49.77)	(112.51)	(43.32)	(48.76)	(62.11)	(114.24)	(85.12)
Sample Size	196,762	273,699	27,319	159,715	196,762	273,699	27,319	159,715	196,762	273,699	27,319	159,715
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Individuals Control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

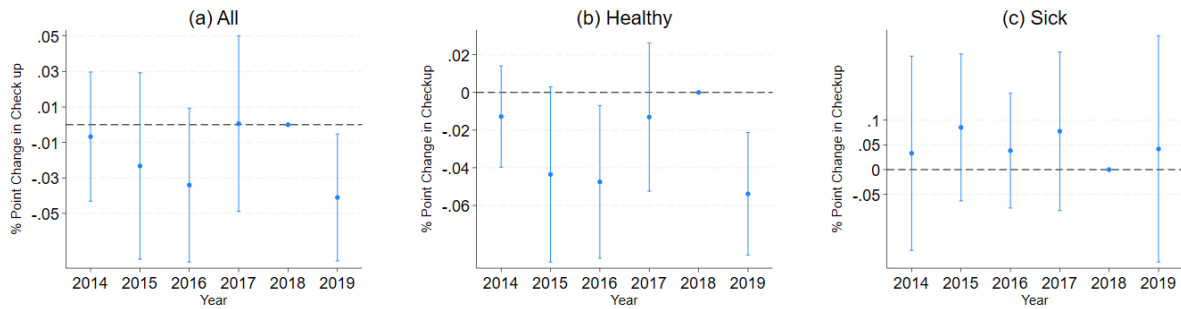
Notes: Unless otherwise stated, the sample consists of adults aged 19 to 64 with household incomes between 138% and 400% of the poverty level (N=189,850). The table presents the coefficient estimate and state-clustered standard error for the interaction between individuals residing in a state that repealed the insurance mandate and premium expenditure in the year 2019. All regressions control for age, sex, marital status, household size, race/ethnicity, educational attainment, employment status, state fixed effects, and year effects. Estimates are weighted using CPS ASEC sampling weights. ***p<0.01, **p<0.05, *p<0.10. Standard errors are in parentheses.

Table D.4: Regression Results for Routine Check Up (Source: BRFSS 2014 to 2019)

	All (1)	Healthy (2)	Sick (3)
Treatment X Post2019	-0.027** (0.011)	-0.030** (0.014)	-0.002 (0.045)
Sample Size	190,208	154,689	35,127
Pre-2019 Mean	0.71	0.70	0.78
Year FE	Yes	Yes	Yes
State FE	Yes	Yes	Yes
Individuals Control	Yes	Yes	Yes

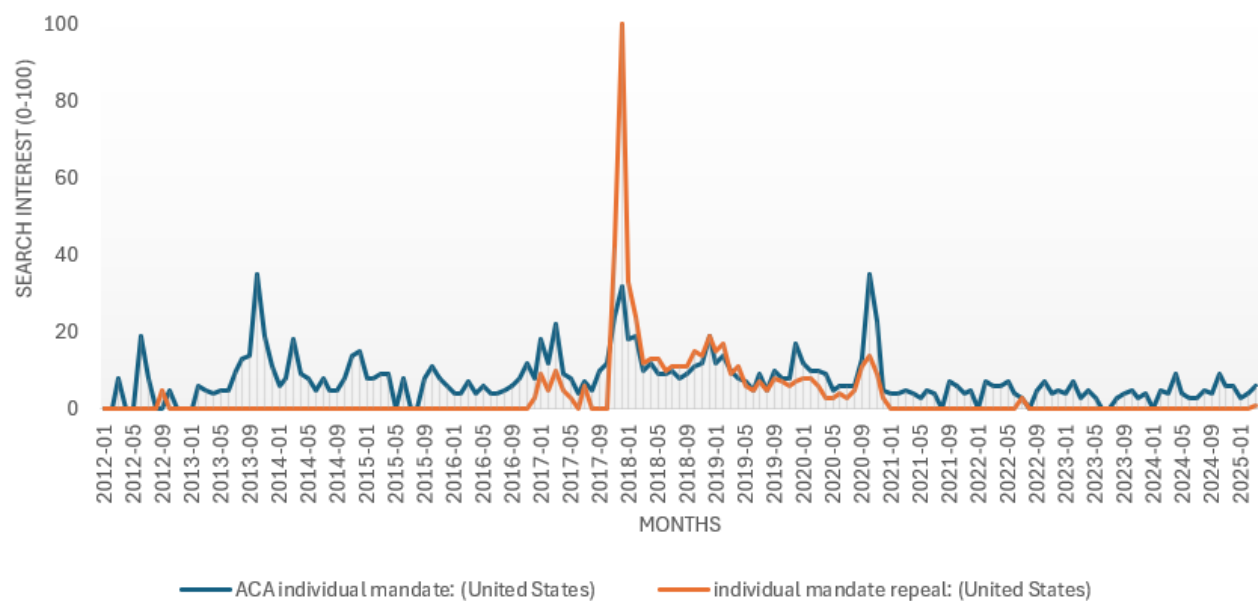
Notes: Sample includes adults aged 19 to 64 years old with household income between 138% and 400% of the federal poverty line (N=190,208). Table displays coefficient estimate and state clustered standard error for the interaction of an indicator for residence in a state that repealed the insurance mandate and the probability of routine check up for full, healthy and sick enrollees. All regressions control for age, sex, marital status, household size, race/ethnicity, educational attainment, employment status, state fixed effects, and year effects. Estimates are weighted by BRFSS sampling weights. ***p<0.01, **p<0.05, *p<0.10. Standard errors are in parentheses.

Figure D.1: Event Study for Routine Check up



Note: Figure 6 illustrates the percentage point changes in routine check-ups before and after the repeal of the individual mandate. Panel (a) illustrates the changes for the full sample. Panel (b) shows the event study for the healthy sample, and Panel (c) depicts the event study plot for the sick sample (Source: Behavioral Risk Surveillance Risk Factor (BRFSS) 2014 to 2019). The sample includes adults aged 19–64 with household income between 138% and 400% of the federal poverty level ($N = 190,708$). The figure presents coefficient estimates with 95% confidence intervals for variables interacting with the mandate-repeal indicator, along with year-specific indicator variables, omitting the 2018 indicator as the reference period. All regressions control for age, sex, marital status, household size, race/ethnicity, educational attainment, employment status, state fixed effects, and year fixed effects. Standard errors are clustered by state, and estimates are weighted using CPS ASEC sampling weights.

Figure D.2: Google Trends on Internet Searches related to ACA Individual Mandate and Mandate Repeal (01/2012-01/2025)



Note: Google Trends data on internet searches related to the ACA individual mandate. (Source: Google Trends; Notes: The figure presents Google Trends data for internet searches of the terms “ACA Individual Mandate” (blue) and “Individual Mandate Repeal” (orange). Search interest for each term is quantified on a scale from 0 to 100, normalized over the selected time period, where 100 represents the peak popularity of that term relative to all other searches during that period.).